AI

NETWORK SECURITY

COURSE SLIDES · WSQ

AI for Network Security

WSQ Course Code: TGS-2024051414

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
Network security, cybersecurity, applied machine learning and secure infrastructure design.

CERTIFICATION
CompTIA Security+ ce · AWS Certified Machine Learning Engineer · AWS Solutions Architect · AWS AI Practitioner.

DELIVERS
WSQ courses on AI for network security, cybersecurity, ethical hacking and IT infrastructure.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Your experience with networking, IT operations or security (if any).
- Any experience you already have with AI or machine-learning tools.
- What you would like to be able to detect, defend or automate after this course.

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HOUSEKEEPING

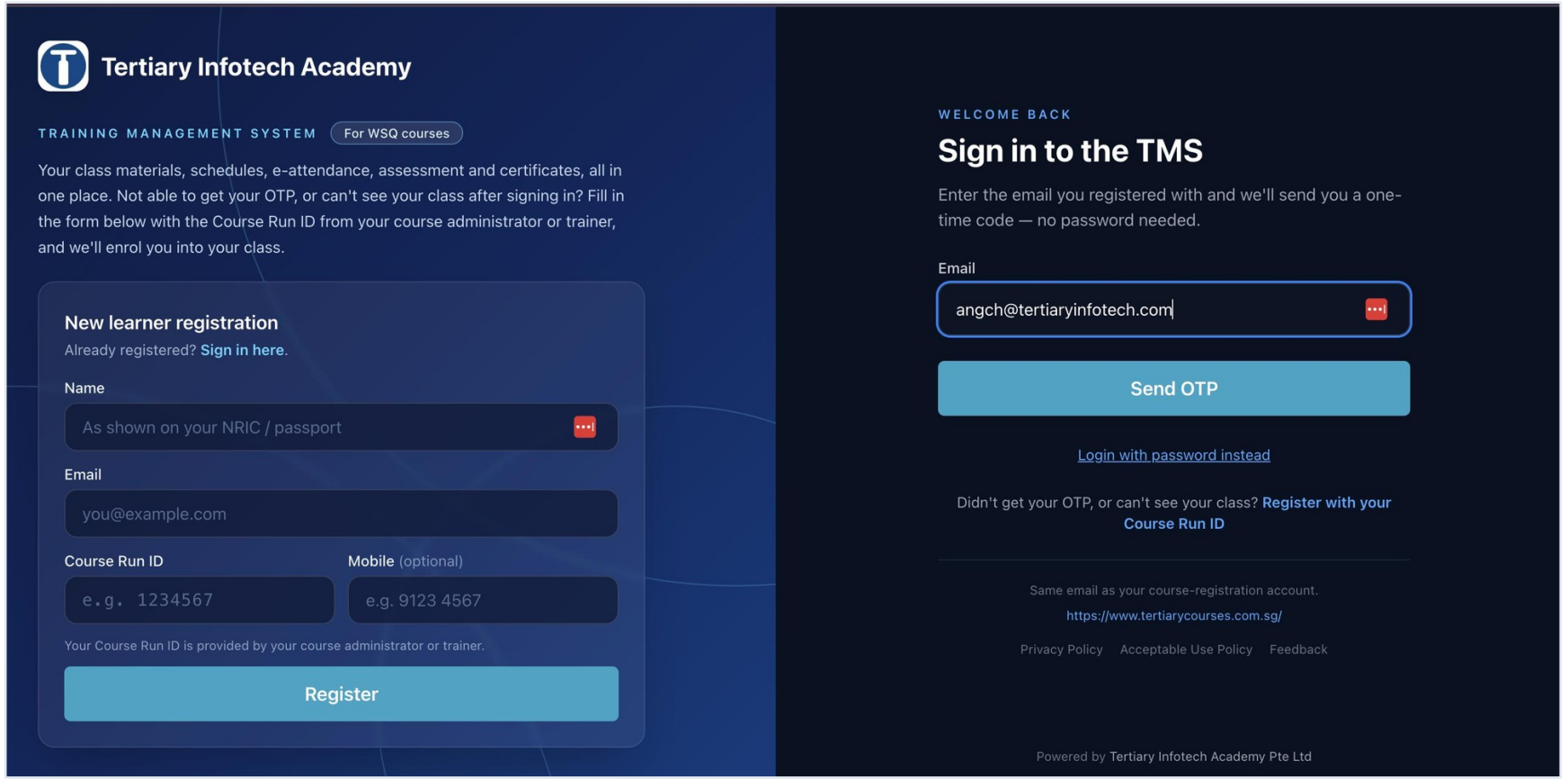
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

Login to LMS and Download Course Material

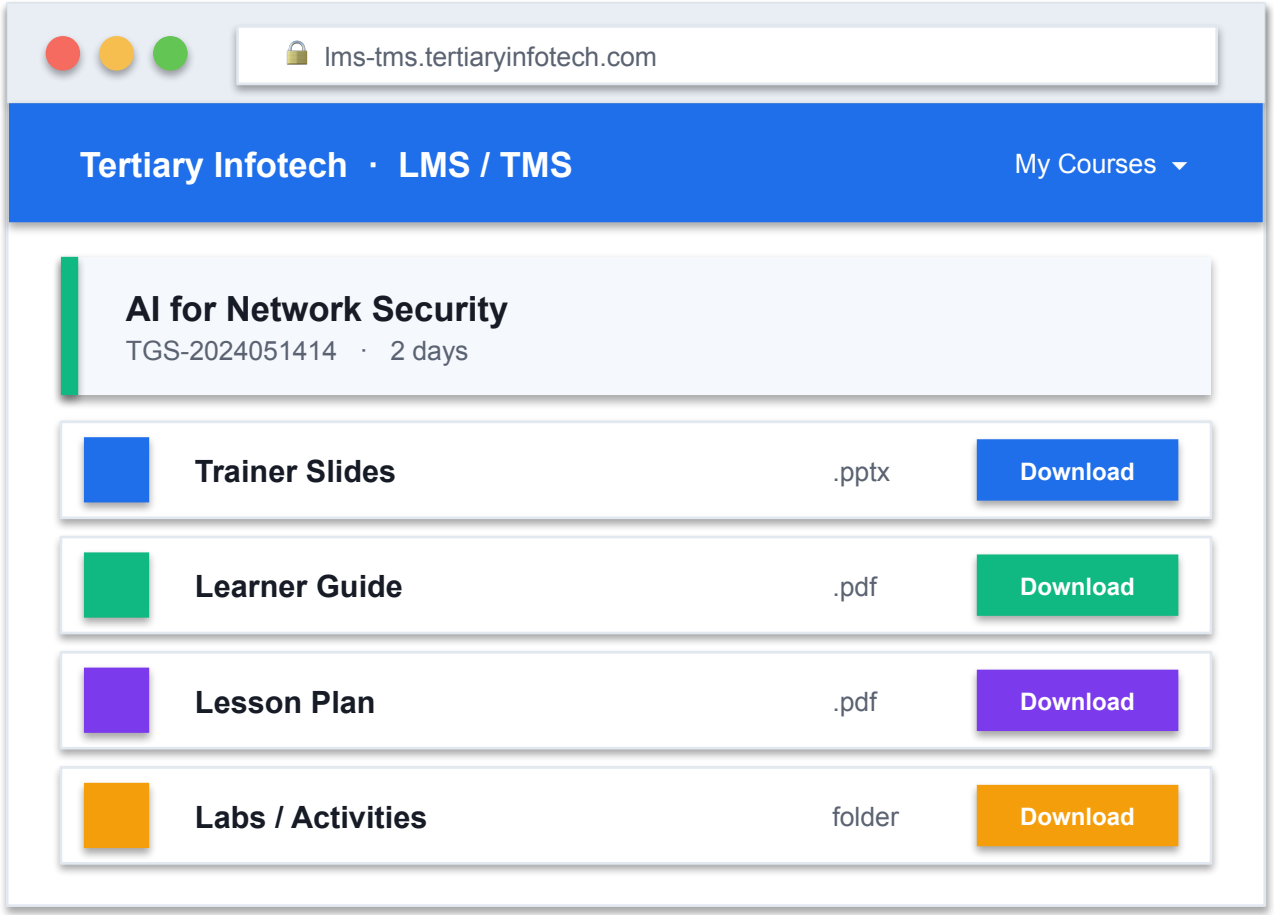


Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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LMS / TMS · lms-tms.tertiaryinfotech.com

Download Your Course Material



1 · Go to the LMS portal

Open lms-tms.tertiaryinfotech.com in your browser.

2 · Log in

Sign in with the e-mail you used to register for this course.

3 · Open this course

Select 'AI for Network Security' from your course list.

4 · Download

Take the Trainer Slides, Learner Guide and Lesson Plan.

5 · Get the labs

Open labs/ — each lab has its own step-by-step guide.

6 · Keep them open

You may use these materials in the open-book assessment.

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Skills Framework Alignment

1

TSC Title

Network Security

2

TSC Code

ICT-DIT-4024-1.1

3

K1 / K2

ICT networks and their configuration; types of network attacks, vulnerabilities and related weaknesses.

4

K3 / K4

Types and techniques of network security measures; risk management plans and procedures.

5

A1 – A4

Identify threats, analyse security risks, determine assets requiring protection and create risk management plans.

6

A5 – A9

Define security phases, develop measures, design auditing and IR procedures, document incidents and implement configurations.

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SCHEDULE · 8 instructional hours per day · 1-hour lunch + 2 tea breaks

Lesson Plan — 2 Days, 8 Training Hours per Day

DAY 1
AI-Assisted Analysis & Risk Planning

9:30	Welcome, learning outcomes, digital attendance (AM)
10:00	Core concepts — network security, the CIA triad, why AI
11:30	Topic 1 — Threat & vulnerability analysis (Labs 1–2)
13:30	Lunch · digital attendance (PM)
14:30	Lab 3 · Topic 2 — Risk assessment & planning
16:30	Topic 2 continued (Labs 4–6) · Day 1 recap

DAY 2
Security Measures & Incident Response

9:30	Day 1 review · digital attendance (AM)
9:45	Topic 3 — Designing AI-enhanced security measures
11:30	Topic 3 continued (Labs 7–9)
13:30	Lunch · digital attendance (PM)
14:30	Topic 4 — Incident documentation & response (Labs 10–11)
16:15	Lab 12 · recap · TRAQOM · assessment attendance
16:45	Briefing for Assessment
17:00	WA-SAQ 1 hr, then Case Study 1 hr

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Threats & vulnerabilities

Analyse ICT configurations with AI assistance to identify network threats and vulnerabilities.

2

LO2 · Assets & risk plans

Evaluate organisational assets and build AI-informed risk management plans for infrastructure weaknesses.

3

LO3 · Security measures

Design and implement AI-enhanced security measures across the planning, building and management phases.

4

LO4 · Incidents & response

Document security incidents and apply incident response configurations using AI-assisted triage.

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FOUR TOPICS · 12 HANDS-ON LABS

Course Outline

1

Topic 01 — AI-Assisted Network Threat and Vulnerability Analysis

ICT networks & configuration · the AI threat landscape · AI anomaly detection · NDR · AI-assisted vulnerability analysis

2

Topic 02 — AI-Powered Network Risk Assessment and Management Planning

Asset discovery & classification · CIA impact · AI risk scoring · risk registers · treatment plans · NIST CSF & AI RMF

3

Topic 03 — Designing and Implementing AI-Enhanced Network Security Measures

Defence in depth · planning/building/management phases · segmentation & zero trust · AI firewalls, IDS/IPS, NAC · AI security itself

4

Topic 04 — AI-Assisted Security Incident Documentation and Response

Logging & syslog · SIEM/SOAR · AI alert triage · the NIST IR lifecycle · incident documentation · playbooks & post-incident review

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BEFORE YOU START

Briefing for Assessment

1

Do · Clear your desk

Place phones and other materials under the table or on the floor.

2

Don't · No recording

No photos or recording of assessment scripts.

3

Don't · No discussion

Work individually — no discussion during the assessment.

4

Do · Black or blue pen

Use a black or blue pen for hard-copy assessments.

5

Don't · No correction fluid

No liquid paper or correction tape may be used.

6

Do · Stop on time

Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

1

Written Assessment (WA-SAQ)

Short-Answer Questions · 1 hour · open book. Tests the underpinning knowledge K1–K4.

2

Case Study (CS)

A realistic organisational scenario · 1 hour · open book. Tests the abilities A1–A9.

3

Open book

You may use the course slides, the Learner Guide and approved materials only.

4

Eligibility

A minimum of 75% attendance is required to be eligible for assessment and funding.

5

Result

You are assessed as Competent (C) or Not Yet Competent (NYC) on each instrument.

6

Appeals

An appeal process is available if you wish to contest an assessment outcome.

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ON ASSESSMENT DAY

Assessment Flow



REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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WSQ FUNDING

Criteria for Funding

1

Attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

Assessment

Complete both assessment components and be assessed as 'Competent'.

3

Digital attendance

Scan the SSG QR code for AM, PM and Assessment on every training day.

4

TRAQOM survey

Complete the mandatory TRAQOM course feedback survey on the LMS.

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LAB ENVIRONMENT

Set Up Before We Start

1

Wireshark

Packet capture and protocol analysis for the traffic-baseline and anomaly labs.

2

Nmap

Host discovery, port and service scanning for the configuration-analysis labs.

3

Python 3 + pandas / scikit-learn

Runs the AI anomaly-detection and log-triage notebooks.

4

A generative AI assistant

Claude, ChatGPT or Copilot — used as the AI analyst copilot throughout the labs.

5

A spreadsheet

Excel, Numbers or Google Sheets for the asset register and risk register labs.

6

A modern browser

For the CVE lookup and the seven browser-based practice tools — no install needed, and the PCAP Analyzer processes your capture locally without uploading it.

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12 LABS · GITHUB REPOSITORY

Access the Hands-On Labs

⌥ https://github.com/tertiarycourses/TGS-2024051414-AI-for-Network-Security

Option A · Clone with Git

- Open a terminal in the folder you want the labs in.
- Run the clone command below.
- The labs/ folder appears with one folder per lab.

```
$ git clone https://github.com/tertiarycourses/TGS-2024051414-AI-for-Network-Security.git
```

Option B · Download the ZIP

- Open the repository link above in your browser.
- Click the green Code button, then Download ZIP.
- Unzip it anywhere on your laptop and open labs/.

EVERY LAB IS SELF-CONTAINED Each labs/lab-NN-*/ folder holds its own step-by-step guide, sample data and expected output. Scan only networks you own or are authorised in writing to test.

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BROWSER TOOLS USED IN THE LABS

Interactive Practice Tools

1

IP Calculator · Labs 1 & 7

Scope the scan and carve one subnet per security zone — alfredang.github.io/ipcalculator

2

PCAP Analyzer · Labs 2 & 11

Read a capture in the browser: protocol mix, top talkers, conversations — alfredang.github.io/pcapanalyzer

3

Cybersecurity Simulator · Lab 5

Run phishing, SQLi and social-engineering attacks to ground your likelihood scores — alfredang.github.io/cybersecuritysimulator

4

Cryptography Toolkit · Labs 8 & 11

AES/DES/RSA and hashing — the confidentiality and integrity controls, and why a hash is tamper-evident — alfredang.github.io/cryptography-toolkit

5

RegexLab · Lab 10

Write and TEST the log filters that drive alert triage, including the negative test — alfredang.github.io/regexgenerator

6

Hacklab · optional extension

A scripted 10-phase penetration-test walkthrough. Not assessed — run it to see what an attacker learns about you — alfredang.github.io/ethnicalhacking

7

FauxBank · optional extension

A safe, deliberately vulnerable banking app with a simulated scanner and report generator. Not assessed — pentest-fauxbank.vercel.app

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CORE CONCEPTS

Network Security, and Why It Now Needs AI

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THE BIG IDEA

What Is Network Security?

1

A combination of controls

The policies, processes and technologies that protect a network and the data crossing it from unauthorised access, misuse and disruption.

2

Layered, not a single tool

Defence in depth: many coordinated layers, so that if one control fails another still stops the attacker.

3

Protects data everywhere

Data at rest, data in transit and data in processing — plus the infrastructure and the people using it.

4

The first line of defence

Detect threats, locate the attack, block access, encrypt the data and keep the network available.

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THE FRAMEWORK BEHIND EVERY CONTROL

The CIA Triad — the Three Goals of Security

1

Confidentiality

Only authorised parties can read the data — delivered by encryption, access control and strong authentication.

2

Integrity

Data is not altered without detection — delivered by cryptographic hashing, digital signatures and validation checks.

3

Availability

Services stay up and reachable — delivered by DDoS protection, redundancy and resilient design.

4

Every control serves the triad

When you evaluate any security control, ask which of the three goals it protects — and at what cost to the others.

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SHARED VOCABULARY

The Language of Networks

1

Host and endpoint

Any device with an address on the network — a laptop, a server, a phone, a camera, a sensor.

2

Switch and router

Switches move frames inside a segment; routers move packets between segments and enforce the boundary.

3

Firewall

The policy enforcement point between zones — it decides which traffic is allowed to cross.

4

Port and service

A service listens on a port; an open port is a door, and every open door is an entry to defend.

5

Protocol

The agreed rules for an exchange — HTTP, DNS, SMB, SSH. Some encrypt by default and some do not.

6

Segment and zone

A group of hosts with a shared trust level; the boundary between zones is where control is applied.

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WHY AI, WHY NOW

Attacks now move at machine speed. Defence cannot stay manual.

Signature matching only finds what someone has already named — and an analyst can only read so many alerts an hour. Artificial intelligence closes both gaps.

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LO1 · THE CORE COMPARISON

Traditional Detection vs AI-Driven Detection

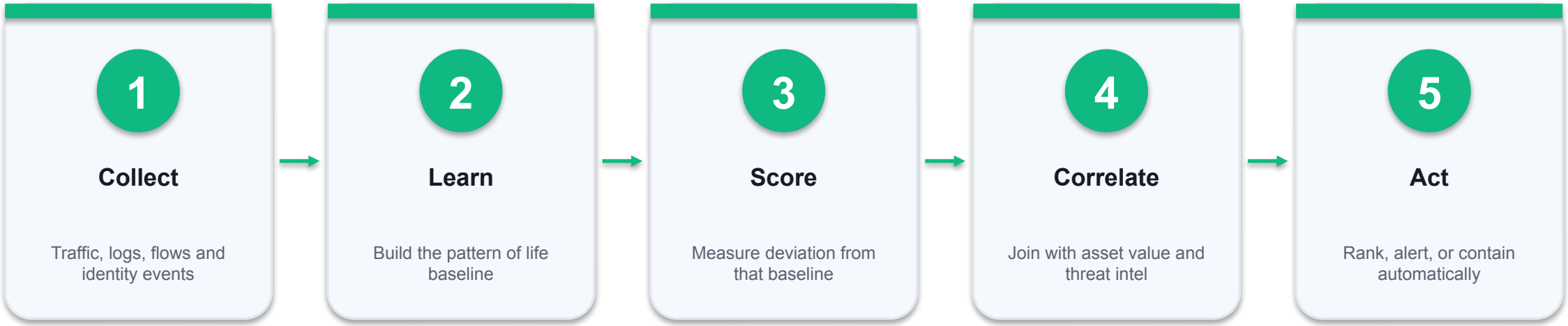
Dimension	Signature-based	AI / behavioural
What it matches	A known pattern in a threat database	A deviation from learned normal behaviour
Zero-day exploits	Structurally cannot detect — no signature exists	Can detect, because the behaviour is still abnormal
Polymorphic malware	Easily evaded by changing the pattern	Harder to evade — the behaviour must also change
False positives	Low on known threats, blind to the rest	Higher initially; falls as the model is tuned
Analyst effort	Manual triage of every alert	Deduplicated, clustered and ranked before a human sees it
Explainability	Excellent — the rule says exactly why	Varies — this is why explainability is a requirement, not a nicety

WHEN IT MATTERS Neither replaces the other. Signatures give precision on named threats; AI gives coverage of the unnamed. A mature operation runs both.

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THE DETECTION PIPELINE

How AI Anomaly Detection Works



THE POINT

The model never learns what an attack is — it learns what normal is, and reports what fails to fit. That is why it can catch a threat nobody has named yet.

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EIGHT APPLICATIONS

Where AI Is Applied in Network Security

1

AI network monitoring

Real-time visibility across hybrid and multi-cloud estates; surfaces misconfigurations and vulnerable assets.

2

Anomaly detection

Learns normal activity and flags deviations in real time — unusual logins, odd hours, unexpected data volumes.

3

AI threat detection

Correlates logs, flows and user behaviour to spot malware, ransomware and multi-stage attacks.

4

Automated response (SOAR)

Isolates a host, revokes a credential or blocks an address in seconds, cutting mean time to remediate.

5

Vulnerability prioritisation

Predicts exploitability and business impact, turning a long scan report into a ranked worklist.

6

Adaptive access control

Scores device, behaviour, location and time, and steps up authentication when risk rises.

7

Phishing detection (NLP)

Reads tone and wording for social-engineering markers; computer vision spots fake login pages.

8

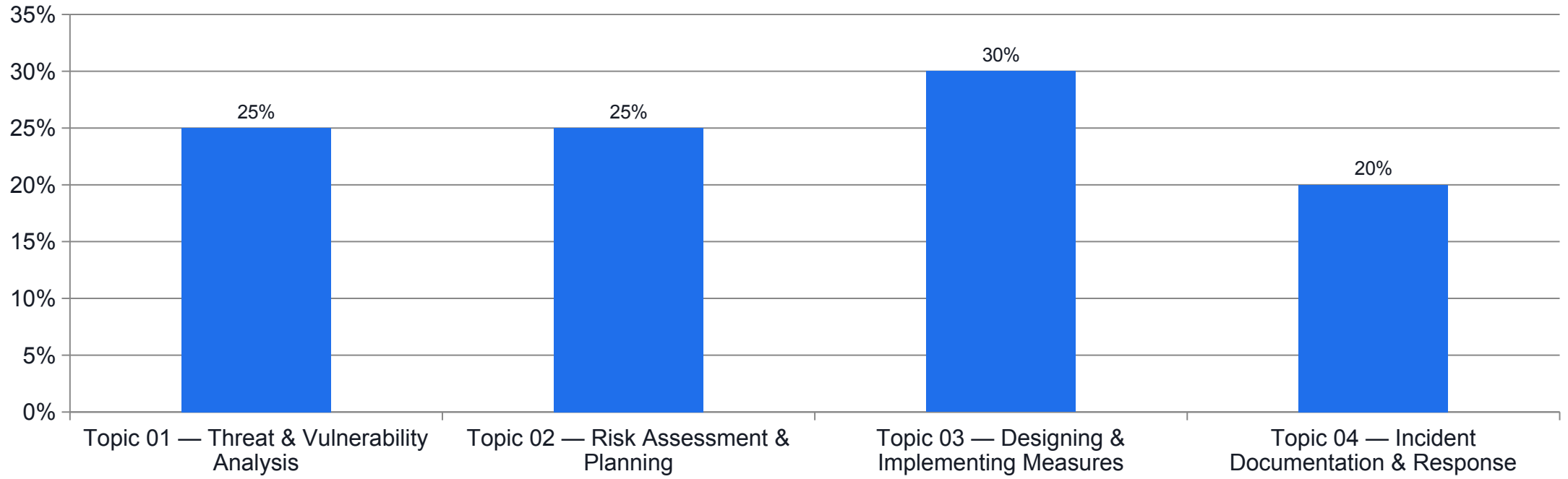
GenAI security copilots

Answer questions in natural language, summarise intelligence and draft incident reports.

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LESSON PLAN · TIME ALLOCATION

Where Course Time Goes — Topic Weighting



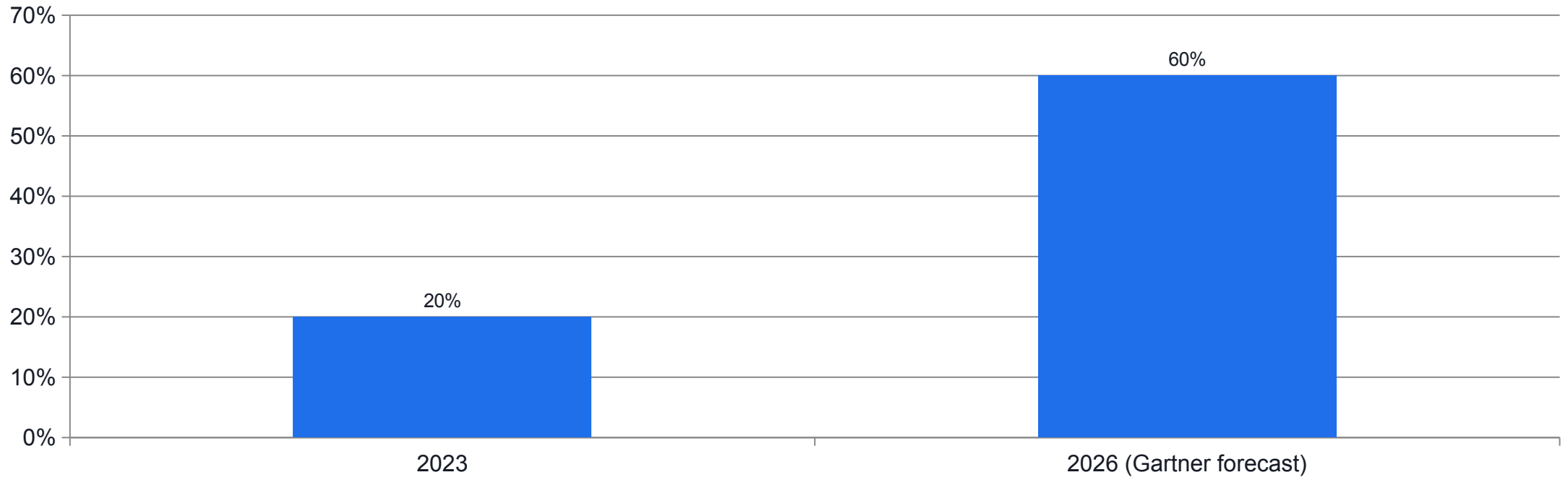
WHAT THE DATA SHOWS

Topic 3 carries the largest weighting because designing and implementing the security measures is where most of the assessable practical ability (A5–A7) sits — plan your revision accordingly.

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THE INDUSTRY SHIFT

Enterprise Adoption of AI-Augmented Security Platforms



WHAT THE DATA SHOWS

Gartner forecasts that over 60% of organisations will rely on cybersecurity platforms with AI-augmented automation in 2026, up from under 20% in 2023 — AI in the SOC has moved from novelty to baseline expectation.

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SETTING EXPECTATIONS HONESTLY

What AI Does NOT Do

1

It does not remove the analyst

AI assists, it does not replace. Context, judgement and accountability stay human.

2

It does not explain itself for free

If you cannot say why the model flagged something, you cannot defend the decision. Demand explainability.

3

It is not immune to attack

Evasion, poisoning, privacy and misuse attacks target the model itself — we cover these in Topic 3.

4

It does not fix bad data

A model trained on incomplete or skewed logs will confidently produce incomplete or skewed results.

5

It does not own the decision

Destructive or wide-blast-radius actions need a named human approver, every time.

6

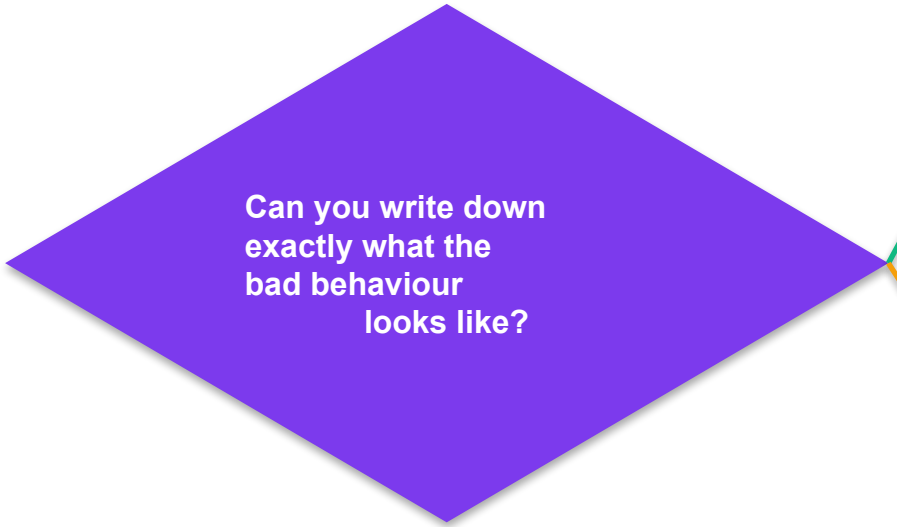
It does not stay accurate by itself

Models drift as the network changes; retraining and revalidation are operational work, not a one-off.

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LO1 · DESIGN DECISION

Should This Detection Be a Rule or a Model?



YES — write a rule

A signature or a threshold rule gives precision, an instant explanation, and no training data. Use it for known, well-defined patterns.

NO — train a model

When 'bad' is only definable as 'not normal', use a behavioural model. Accept that it needs a baseline, tuning and explainability work.

Most real detections need both: the rule catches the named threat cheaply, the model covers the space of threats nobody has named yet.

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01

TOPIC 01

AI-Assisted Network Threat and Vulnerability Analysis

ICT networks & configuration · the AI threat landscape · AI anomaly detection · NDR · AI-assisted vulnerability analysis

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Topic 01 — What It Covers and How It Is Assessed

1

Learning outcome

LO1 — Identify and assess network threats and vulnerabilities by analysing ICT configurations.

2

Knowledge assessed (WA-SAQ)

K1: Infocomm Technology (ICT) networks and their configuration

3

Abilities assessed (Case Study)

A1: Identify threats to network security · A2: Analyse security risks

4

Hands-on labs

Labs 1, 2, 3 — 25% of course time.

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TOPIC WEIGHTING 25%

Key Concepts — Topic 01

1. An ICT network is a layered system — hosts, switches, routers, firewalls, wireless, servers and cloud edges — and every layer carries its own configuration and its own attack surface.

2. Traditional signature-based detection fails against zero-day exploits, polymorphic malware and Living-off-the-Land (LOTL) techniques; AI closes the gap by learning behaviour rather than matching known strings.

3. AI network monitoring gives real-time visibility across hybrid and multi-cloud estates, flags misconfigurations and surfaces vulnerable assets before an incident escalates.

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TOPIC WEIGHTING 25%

Key Concepts — Topic 01 (continued)

1. AI anomaly detection learns the 'pattern of life' of a network — who normally talks to whom, when, and how much — then reports statistically significant deviations as candidate intrusions.

2. Network Detection and Response (NDR) applies behavioural AI to reconnaissance, credential abuse, lateral movement and data staging across the whole attack lifecycle.

3. AI-assisted vulnerability analysis adds contextual prioritisation to a raw scan: predicted exploitability and business impact turn a 900-line report into a ranked worklist.

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The Threat Landscape You Are Defending Against

1 **Malware & ransomware**
Code that encrypts, steals or destroys — increasingly polymorphic, so its signature changes every build.

2 **Phishing & social engineering**
The most common initial access route; AI-generated lures are now fluent and personalised.

3 **DDoS**
Floods a service until it is unavailable — an attack on the availability leg of the CIA triad.

4 **Man-in-the-middle**
Intercepts or alters traffic in transit; defeated by strong encryption and certificate validation.

5 **Advanced Persistent Threats**
Patient, well-resourced intruders who dwell for months — behaviour analytics is the main way to find them.

6 **Living off the Land (LOTL)**
Uses the built-in admin tools already on the host, so there is no malware file to detect.

7 **Insider threat**
An authorised user acting against the organisation — only detectable as a deviation from their own norm.

8 **AI-enabled attacks**
Adversaries use AI to raise the speed, scale and quality of attacks — defence must match that tempo.

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Building a 'Pattern of Life' Baseline



WHY PER-ENTITY

A backup server moving 400 GB at 02:00 is normal; a laptop doing it is not. A single global threshold cannot express that — a per-entity baseline can.

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Network Detection and Response (NDR)

1

What it is

A platform that applies behavioural AI to network traffic to detect attacker activity that endpoint and perimeter tools miss.

2

Reconnaissance

Spots scanning and enumeration — one host suddenly touching many hosts or many ports.

3

Credential abuse

Flags an account used from a new device, a new location or at an impossible hour.

4

Lateral movement

Detects east-west traffic that crosses a zone boundary it has never crossed before.

5

Data staging & exfiltration

Notices collection into an unusual location and outbound volume that breaks the baseline.

6

Encrypted traffic

Infers behaviour from flow metadata — timing, volume and direction — without needing to decrypt.

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Prioritising Vulnerabilities — Three Approaches

Approach	How it ranks	Weakness	Verdict
CVSS only	By base severity score	Ignores whether it is exploited and what the asset is worth	Necessary, not sufficient
CVSS + KEV	Severity, with known-exploited first	Still ignores business context	Much better — free and fast
AI-assisted contextual	Predicted exploitability x asset criticality x exposure	Needs a trustworthy asset register	Best — and it is what Lab 3 builds

WHEN IT MATTERS A ranking you cannot explain is a ranking nobody will fund. Whatever method you choose, write the formula down so the order is reproducible.

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WHAT YOU'LL DO

Hands-On Labs — Topic 01

Lab 1

- Map the ICT Network and Analyse Its Configuration
-
- Establish what the network actually is before judging whether it is safe: discover live hosts, open ports and services,

Lab 2

- Baseline Normal Traffic and Detect an Anomaly with AI
-
- Learn what 'normal' looks like on a network, then let a machine-learning model tell you when traffic stops being normal.

Lab 3

- AI-Assisted Vulnerability Analysis and Prioritisation
-
- Turn a raw vulnerability list into a ranked, defensible worklist by combining CVSS severity with exploitability and busi

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TOPIC 01 · HANDS-ON

Map the ICT Network and Analyse Its Configuration

LAB 1

K1 · A1 — Establish what the network actually is before judging whether it is safe: discover live hosts, open ports and services, and read the configuration for weaknesses.

You are handed an unfamiliar subnet and asked the question every security review starts with: what is on this network, and what is it running? You use Nmap to discover live hosts and their open ports and service versions, record the result as a structured inventory, and then read the configuration evidence for the weaknesses that matter — management ports exposed, unencrypted protocols in use, out-of-date service versions and default configurations left in place.

YOU'LL BUILD

A network configuration inventory (host, IP, OS guess, open ports, service versions) plus a written list of the configuration weaknesses you found and why each one matters.

TOOLCHAIN

Nmap, Zenmap, the IP Calculator (alfredang.github.io/ipcalculator), a spreadsheet, a generative AI assistant

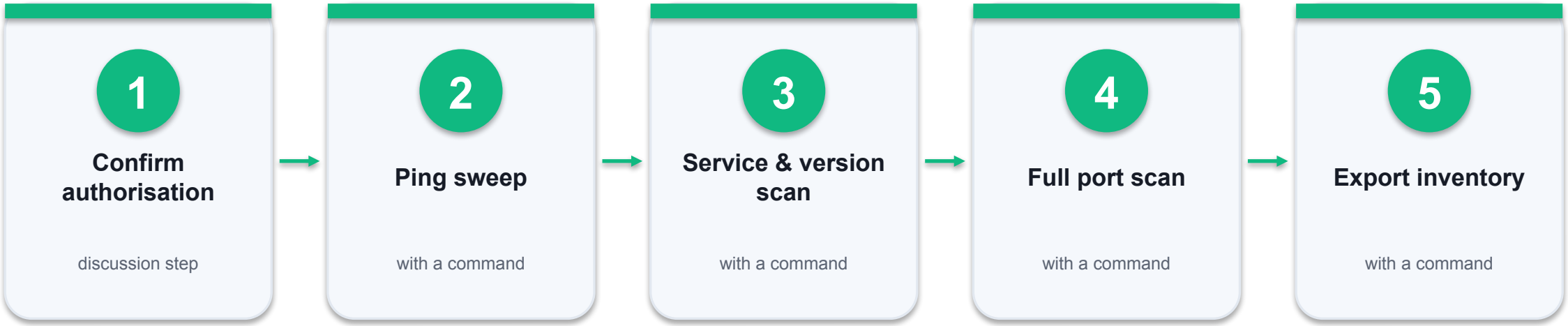
DONE WHEN

Your inventory names every live host with its open ports and service versions, and your weakness list contains at least five configuration weaknesses, each tied to specific evidence in the scan output. You can state, for each AI-suggested finding, whether the scan evidence actually supports it.

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LAB 1 · PROCESS MAP

How Lab 1 Runs



THE POINT

K1 · A1 — Establish what the network actually is before judging whether it is safe: discover live hosts, open ports and services, and read the configuration for weaknesses.

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LAB 1 · PROCEDURE

Map the ICT Network and Analyse Its Configuration — Part 1/2

1. Confirm you are scanning only the lab network you have been authorised to scan. Never scan a network you do not own or have written permission to test.

2. **Before you scan anything, work out exactly what 192.168.56.0/24 covers so your scan scope is defensible. Open the IP Calculator, enter the CIDR, and record the network address, the broadcast address, the usable host range and the host count.**

```
→ https://alfredang.github.io/ipcalculator/
```

3. **Discover which hosts are alive on the lab subnet with a ping sweep. Note how many hosts answer, and compare that against the usable host count you just calculated.**

```
$ nmap -sn 192.168.56.0/24
```

4. **Run a service and version scan against the live hosts to learn what each one is actually running.**

```
$ nmap -sV -O -T4 192.168.56.0/24 -oN inventory.txt
```

5. **Scan the full TCP port range on the one host that looks most interesting, so nothing above port 1024 is missed.**

```
$ nmap -sV -p- -T4 192.168.56.101 -oN host101-full.txt
```


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LAB 1 · PROCEDURE

Map the ICT Network and Analyse Its Configuration — Part 2/2

6 **Export the scan to a machine-readable form so it can be loaded into a spreadsheet or a script.**

```
$ nmap -sV 192.168.56.0/24 -oX inventory.xml
```

7 Build the inventory table in a spreadsheet with the columns Host, IP, OS, Open Ports, Services, Versions, Owner, Business Function.

8 Read the inventory for configuration weaknesses. Flag every unencrypted management protocol (Telnet 23, FTP 21, HTTP 80 on an admin interface), every service version that is behind, and every port that has no business being open.

9 **Paste the sanitised Nmap output into your AI assistant and ask it to identify the configuration weaknesses and explain the risk of each. Compare its list against yours.**

```
$ Prompt: 'Here is Nmap output from a network I am authorised to assess. Identify configuration weaknesses, name the affected protocol or service, and explain the risk of each in one sentence.'
```

10 Reconcile the two lists. Record anything the AI found that you missed, and anything it claimed that the evidence does not support — the second list is the more important one.

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LAB 1 · VERIFY

Map the ICT Network and Analyse Its Configuration

✅ Expected result

Your inventory names every live host with its open ports and service versions, and your weakness list contains at least five configuration weaknesses, each tied to specific evidence in the scan output. You can state, for each AI-suggested finding, whether the scan evidence actually supports it.

IF IT DOESN'T WORK

No hosts answer the ping sweep

the lab hosts may block ICMP; re-run with `nmap -Pn`.

The version scan is slow

narrow the port range with `-p 1-1024` or raise the timing to `-T4`.

`-O` reports 'too many fingerprints'

OS detection needs at least one open and one closed port; it is a hint, not evidence.

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TOPIC 01 · HANDS-ON

Baseline Normal Traffic and Detect an Anomaly with AI

LAB 2

K1 · K2 · A1 — Learn what 'normal' looks like on a network, then let a machine-learning model tell you when traffic stops being normal.

Signature detection only finds what someone has already named. This lab builds the other half of the picture. You capture live traffic in Wireshark, turn it into per-connection features, train an unsupervised anomaly-detection model on the quiet 'pattern of life' period, and then score a period that contains a port scan and a data-staging burst. You see for yourself which events the model scores as anomalous and — just as importantly — which normal events it wrongly flags.

YOU'LL BUILD

A labelled anomaly report: the connections your Isolation Forest model scored as anomalous, ranked by score, with your own verdict of true positive or false positive against the packet evidence.

TOOLCHAIN

Wireshark, tshark, the PCAP Analyzer (alfredang.github.io/pcapanalyzer), Python 3, pandas, scikit-learn, a generative AI assistant

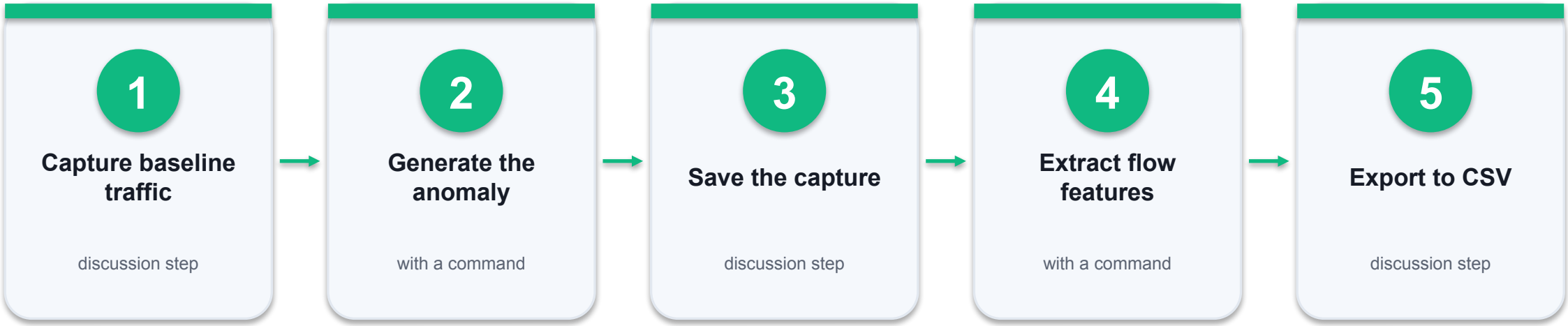
DONE WHEN

The port scan appears in your top-ranked anomalies — it shows as many short connections from one source to many destination ports. You can name at least one false positive and explain why the model flagged legitimate traffic, and you can state your model's precision on the twenty you reviewed.

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LAB 2 · PROCESS MAP

How Lab 2 Runs



THE POINT

K1 · K2 · A1 — Learn what 'normal' looks like on a network, then let a machine-learning model tell you when traffic stops being normal.

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LAB 2 · PROCEDURE

Baseline Normal Traffic and Detect an Anomaly with AI — Part 1/3

1. Start a Wireshark capture on the lab interface and leave it running while you browse and use the lab services normally for five minutes. This is your baseline period.

2. **While the capture runs, generate the anomalous period: run a port scan against a lab host and then copy a large file across the network.**

```
$ nmap -sS -p- 192.168.56.101
```

3. Stop the capture and save it as traffic.pcapng.

4. **Before writing any code, get a fast read on what you captured. Drag traffic.pcapng into the PCAP Analyzer (it runs entirely in your browser — nothing is uploaded) and note the packet count, capture duration, protocol distribution, top talkers and top conversations.**

```
→ https://alfredang.github.io/pcapanalyzer/
```


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LAB 2 · PROCEDURE

Baseline Normal Traffic and Detect an Anomaly with AI — Part 2/3

5 Write down which conversation you EXPECT the model to flag as anomalous, based on the top-talkers view alone. You will check that prediction against the model's output in step 10 — a human hypothesis first is what stops you rationalising whatever the model happens to say.

6 **Export per-connection features from the capture with tshark — source, destination, port, packet count, byte count and duration.**

```
$ tshark -r traffic.pcapng -q -z conv,tcp > conversations.txt
```

7 **Export the same data as CSV for the model to read.**

```
$ tshark -r traffic.pcapng -T fields -e ip.src -e ip.dst -e tcp.dstport -e frame.len -e frame.time_relative -E header=y -E separator=, > traffic.csv
```

8 **Load traffic.csv into pandas and aggregate it per (source, destination, port) conversation: packet count, total bytes, mean packet size and duration.**

```
$ python3 baseline.py   # provided in labs/lab-02/
```


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LAB 2 · PROCEDURE

Baseline Normal Traffic and Detect an Anomaly with AI — Part 3/3

9 **Fit an Isolation Forest on the baseline period only, with a contamination rate of 0.02, then score every conversation in the full capture.**

```
$ from sklearn.ensemble import IsolationForest
```

10 **Sort by score ascending and inspect the twenty most anomalous conversations. Open each one in Wireshark using a display filter and decide what it actually was.**

```
$ ip.addr == 192.168.56.101 && tcp.port == 445
```

11 Mark each of the twenty as a true positive or a false positive, and write one sentence saying what evidence decided it.

12 **Ask your AI assistant to explain the anomalies and suggest which features would separate the scan from the file copy more cleanly. Test one of its suggestions and record whether it improved the result.**

```
$ Prompt: 'These network conversations were scored as anomalous by an Isolation Forest. Given packet count, byte count and duration, explain what behaviour each likely represents and suggest one additional feature that would separate port scanning from a legitimate large file transfer.'
```


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LAB 2 · VERIFY

Baseline Normal Traffic and Detect an Anomaly with AI

✅ Expected result

The port scan appears in your top-ranked anomalies — it shows as many short connections from one source to many destination ports. You can name at least one false positive and explain why the model flagged legitimate traffic, and you can state your model's precision on the twenty you reviewed.

IF IT DOESN'T WORK

tshark is not found

it ships with Wireshark; on macOS add /Applications/Wireshark.app/Contents/MacOS to PATH.

The capture is empty

you may be capturing on the wrong interface; list them with `tshark -D`.

Everything scores as anomalous

your baseline period contained the attack traffic; re-cut the baseline to the quiet window only.

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TOPIC 01 · HANDS-ON

AI-Assisted Vulnerability Analysis and Prioritisation

LAB 3

K2 · A1 · A2 — Turn a raw vulnerability list into a ranked, defensible worklist by combining CVSS severity with exploitability and business context.

A scanner will hand you hundreds of findings and no opinion about which one to fix on Monday morning. In this lab you take the service inventory from Lab 1, look up the real CVEs affecting those service versions, and then build a prioritisation that a manager would actually accept: CVSS base score combined with known-exploited status and the business criticality of the affected asset. You use an AI assistant to accelerate the research and then verify every claim it makes against the authoritative source.

YOU'LL BUILD

A prioritised vulnerability worklist: CVE, affected asset, CVSS score, known-exploited flag, asset criticality, computed priority and a one-line recommended fix — ranked, with the top five justified.

TOOLCHAIN

Nmap service inventory, NVD, CISA KEV catalogue, a spreadsheet, a generative AI assistant

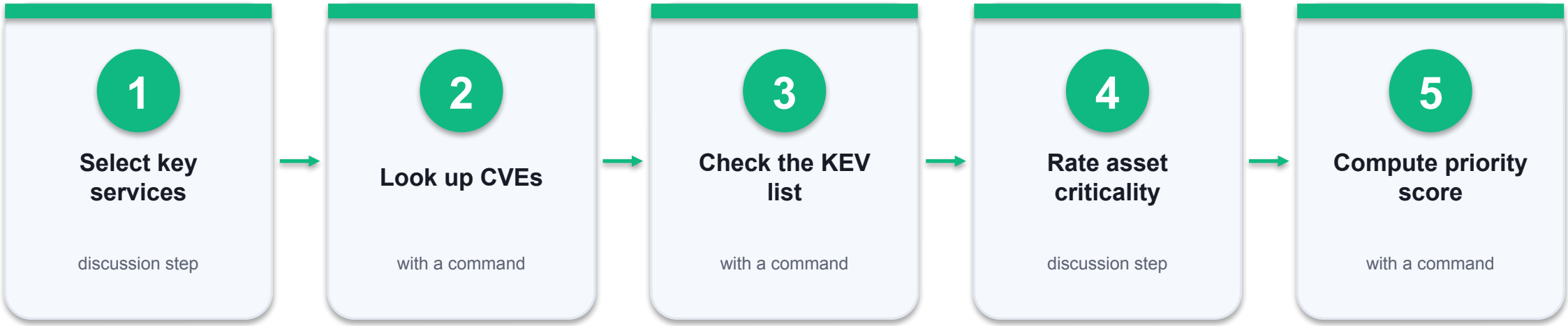
DONE WHEN

Your worklist ranks every finding by a computed priority, not by CVSS alone; a KEV-listed CVE on a high-criticality asset sits above a higher-CVSS finding on a low-criticality one. You can point to at least one AI-supplied detail you corrected against the NVD.

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LAB 3 · PROCESS MAP

How Lab 3 Runs



THE POINT

K2 · A1 · A2 — Turn a raw vulnerability list into a ranked, defensible worklist by combining CVSS severity with exploitability and business context.

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LAB 3 · PROCEDURE

AI-Assisted Vulnerability Analysis and Prioritisation — Part 1/2

1. Take the service-version list from Lab 1. Pick the six most significant services — anything internet-facing, anything running as an administrator, anything holding data.

2. **For each service version, search the NIST National Vulnerability Database for matching CVEs and record the CVE ID and CVSS v3 base score.**

```
→ https://nvd.nist.gov/vuln/search
```

3. **Check each CVE against the CISA Known Exploited Vulnerabilities catalogue. A CVE on the KEV list is being exploited in the wild right now and outranks a higher CVSS score that is not.**

```
→ https://www.cisa.gov/known-exploited-vulnerabilities-catalog
```

4. Assign each affected asset a business criticality of High, Medium or Low, based on what a loss of confidentiality, integrity or availability would cost. Write the justification in the row.

5. **Compute a priority score for every finding so the ranking is reproducible rather than a matter of taste.**

```
$ Priority = CVSS x Criticality weight x KEV multiplier
```


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LAB 3 · PROCEDURE

AI-Assisted Vulnerability Analysis and Prioritisation — Part 2/2

6 Sort by priority and inspect the top ten. Check that the order matches your professional judgement — if it does not, your weights are wrong, so fix the weights rather than the ranking.

7 **Ask your AI assistant to summarise each of your top five CVEs and propose a remediation for each.**

```
$ Prompt: 'For each of these CVEs, give the vulnerability class, the attack prerequisite, the practical impact, and the recommended remediation. State the CVSS v3 base score. Flag any CVE you are not certain about.'
```

8 Verify every AI-supplied CVSS score and remediation against the NVD entry. Correct anything wrong and note where the AI was inaccurate — this is the evidence that you validated the model rather than trusted it.

9 Write the two-sentence justification for your number-one priority: what could happen, and why it outranks everything else on the list.

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LAB 3 · VERIFY

AI-Assisted Vulnerability Analysis and Prioritisation

✅ Expected result

Your worklist ranks every finding by a computed priority, not by CVSS alone; a KEV-listed CVE on a high-criticality asset sits above a higher-CVSS finding on a low-criticality one. You can point to at least one AI-supplied detail you corrected against the NVD.

IF IT DOESN'T WORK

NVD returns no CVE for a version string

search the product name and major version rather than the full banner.

Every finding scores the same

your criticality ratings are all identical; force a spread by ranking assets against each other.

The AI invents a CVE ID

this is exactly why step 8 exists; a CVE that is not in the NVD does not exist.

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TOPIC RECAP

Recap — Topic 01

1

Lab 1 — Map the ICT Network and Analyse Its Configuration

Establish what the network actually is before judging whether it is safe: discover live hosts, open ports and services, and read the configuration for weaknesses.

2

Lab 2 — Baseline Normal Traffic and Detect an Anomaly with AI

Learn what 'normal' looks like on a network, then let a machine-learning model tell you when traffic stops being normal.

3

Lab 3 — AI-Assisted Vulnerability Analysis and Prioritisation

Turn a raw vulnerability list into a ranked, defensible worklist by combining CVSS severity with exploitability and business context.

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02

TOPIC 02

AI-Powered Network Risk Assessment and Management Planning

Asset discovery & classification · CIA impact · AI risk scoring · risk registers · treatment plans · NIST CSF & AI RMF

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TOPIC 02 · ROADMAP

Topic 02 — What It Covers and How It Is Assessed

1

Learning outcome

LO2 — Evaluate organisational assets and develop risk management plans to address network infrastructure weaknesses effectively.

2

Knowledge assessed (WA-SAQ)

K2: Types of network attacks, vulnerabilities and related weaknesses of installed infrastructure

3

Abilities assessed (Case Study)

A3: Determine organisational assets that require protection · A4: Create risk management plans to mitigate risks

4

Hands-on labs

Labs 4, 5, 6 — 25% of course time.

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TOPIC WEIGHTING 25%

Key Concepts — Topic 02

1. You cannot protect what you have not inventoried — AI-driven asset discovery continuously fingerprints every identity, device and workload rather than relying on a stale spreadsheet.

2. Asset criticality is a business judgement expressed against the CIA triad: what does a loss of confidentiality, integrity or availability actually cost this organisation?

3. Risk = Likelihood x Impact. AI improves the likelihood term with exploit-prediction models and threat-intelligence correlation, and improves the impact term with real dependency mapping.

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TOPIC WEIGHTING 25%

Key Concepts — Topic 02 (continued)

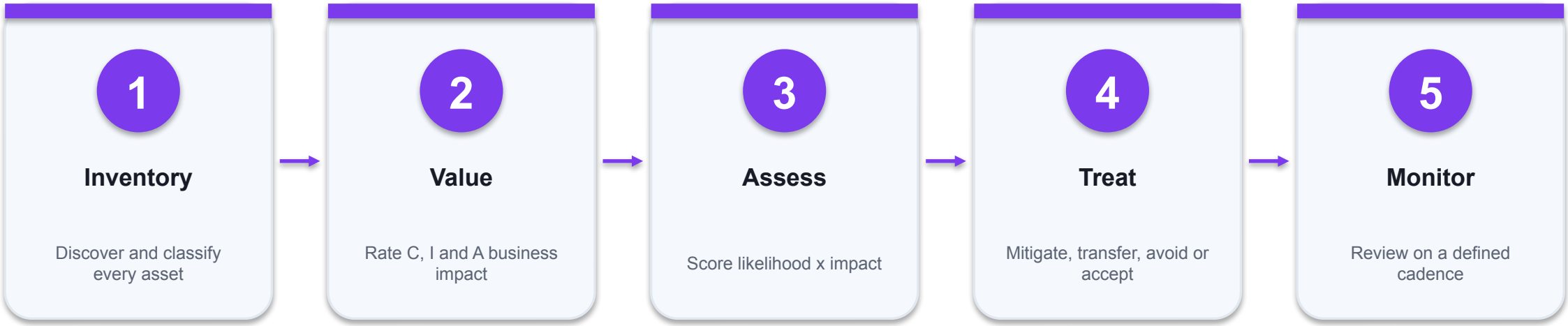
1. A risk register is the working artefact of risk management: asset, threat, vulnerability, existing control, inherent risk, treatment, owner, target date and residual risk.

2. The four treatment options are mitigate, transfer, avoid and accept — every entry on the register must land on one of them, with a named owner.

3. The NIST Cybersecurity Framework (Govern, Identify, Protect, Detect, Respond, Recover) and the NIST AI Risk Management Framework (Govern, Map, Measure, Manage) give the planning scaffold.

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From Asset to Treated Risk



THE DISCIPLINE

Every step produces a named artefact — asset register, impact rating, risk register, treatment plan, review schedule. If a step has no artefact, it did not happen.

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LO2 · A2 · SCORING CONSISTENTLY

The 5x5 Risk Matrix — Reading the Score

Impact ↓ / Likelihood →	1 Rare	2 Unlikely	3 Possible	4 Likely	5 Almost certain
5 · Severe	5 Med	10 High	15 Critical	20 Critical	25 Critical
4 · Major	4 Low	8 Med	12 High	16 Critical	20 Critical
3 · Moderate	3 Low	6 Med	9 Med	12 High	15 Critical
2 · Minor	2 Low	4 Low	6 Med	8 Med	10 High
1 · Negligible	1 Low	2 Low	3 Low	4 Low	5 Med

WHEN IT MATTERS Bands: 1–4 Low · 5–9 Medium · 10–14 High · 15–25 Critical. Agree the bands before you score, or every reviewer will argue the colour instead of the risk.

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The Four Risk Treatments

1

Mitigate

Reduce likelihood or impact with a control. The default choice — and the one that costs money.

2

Transfer

Shift the financial consequence to someone else, typically insurance or a contractual clause. The risk still happens.

3

Avoid

Stop doing the activity that creates the risk — decommission the service, drop the feature, close the port.

4

Accept

Consciously carry the risk, with a named accepting authority and a written rationale. Not the same as ignoring it.

5

Every risk gets exactly one

A blank treatment column is an unmanaged risk wearing a spreadsheet.

6

Residual risk is what remains

After the treatment is applied. If residual still exceeds appetite, the treatment was not enough.

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Two Frameworks You Will Use Together

	NIST Cybersecurity Framework 2.0	NIST AI Risk Management Framework
Protects	The network and the organisation	The AI system itself
Functions	Govern · Identify · Protect · Detect · Respond · Recover	Govern · Map · Measure · Manage
You use it to	Structure the control set and find coverage gaps	Structure how you assess and manage AI risk
In this course	Lab 6 maps your plan to the six functions	Lab 9 hardens the AI you deployed
Why both	Deploying AI to defend the network adds a second thing to defend — the AI	Neither framework covers the other's scope

WHEN IT MATTERS A plan that secures the network with AI but never assesses the AI's own risk has simply moved the weakest link, not removed it.

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WHAT YOU'LL DO

Hands-On Labs — Topic 02

Lab 4

- Build an Asset Register and Classify What Needs Protecting
-
- Determine which organisational assets require protection, and defend the classification with a business impact judgement

Lab 5

- AI-Assisted Risk Scoring and the Risk Register
-
- Combine the asset register and the vulnerability worklist into a scored risk register, using AI to sharpen the likelihoo

Lab 6

- Develop the Risk Management Plan
-
- Turn the scored register into a risk management plan with treatments, owners, dates and a monitoring cadence that surviv

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TOPIC 02 · HANDS-ON

Build an Asset Register and Classify What Needs Protecting

LAB 4

A3 · K1 — Determine which organisational assets require protection, and defend the classification with a business impact judgement rather than a gut feeling.

Risk management starts with an honest asset register. In this lab you take the network inventory from Lab 1 and turn it into an organisational asset register: you add the owner, the data the asset holds, the business function it serves, and a classification against confidentiality, integrity and availability. You then rank the assets by criticality and use an AI assistant to challenge your classification — because the asset you forgot is the one that gets you.

YOU'LL BUILD

A classified asset register with an owner, data classification and C/I/A impact rating for every asset, ranked by criticality, plus the crown-jewel shortlist.

TOOLCHAIN

Lab 1 inventory, a spreadsheet, a generative AI assistant

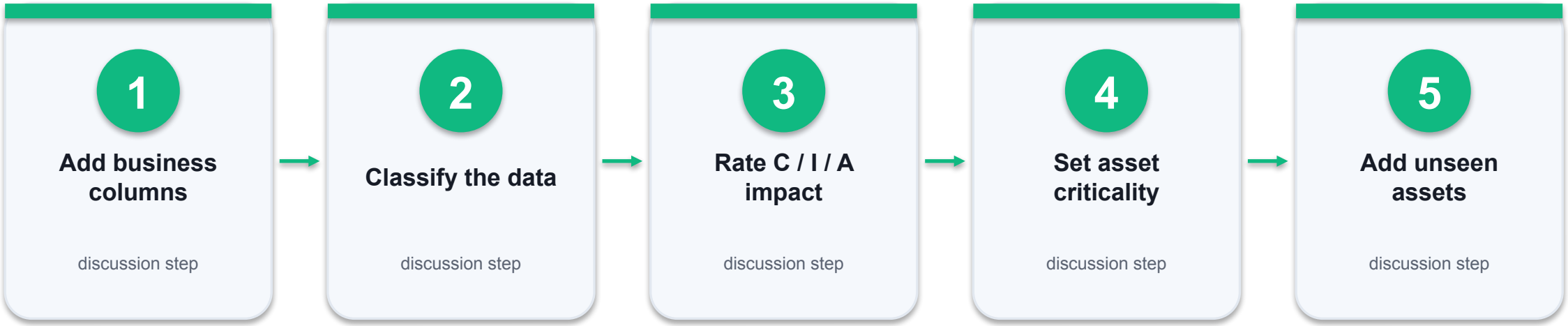
DONE WHEN

Every row has an owner, a data classification with a reason, three C/I/A ratings and a criticality. Your register includes at least two non-technical assets that no scanner would have found, and you can name your crown jewels and say what makes each one critical.

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LAB 4 · PROCESS MAP

How Lab 4 Runs



THE POINT

A3 · K1 — Determine which organisational assets require protection, and defend the classification with a business impact judgement rather than a gut feeling.

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LAB 4 · PROCEDURE

Build an Asset Register and Classify What Needs Protecting — Part 1/2

1. Open the network inventory from Lab 1 and add the organisational columns: Asset Name, Owner, Business Function, Data Held, Data Classification, Confidentiality Impact, Integrity Impact, Availability Impact, Criticality.
2. Classify the data each asset holds as Public, Internal, Confidential or Restricted. Write the reason in the row — an unjustified classification cannot be defended in a review.
3. Rate the Confidentiality, Integrity and Availability impact of losing each asset as High, Medium or Low. Ask the business question, not the technical one: what would this cost the organisation?
4. Set the asset Criticality to the highest of its three C/I/A ratings. An asset with High availability impact is critical even if its data is public — a payment gateway is the obvious case.
5. Add the assets that a network scan can never see: the people, the paper processes, the third-party services, the backups and the domain registration. These are real assets with real impact.

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LAB 4 · PROCEDURE

Build an Asset Register and Classify What Needs Protecting — Part 2/2

6 Sort by criticality and identify the crown jewels — the three to five assets whose loss would genuinely damage the organisation.

7 **Ask your AI assistant to review your register for gaps.**

```
$ Prompt: 'Here is an asset register for a small enterprise network. Identify asset categories I have likely missed, and challenge any criticality rating that looks inconsistent with the stated business function. Do not rewrite my register — list the gaps and the inconsistencies.'
```

8 Act on the review: add every legitimate missing asset, and either correct or justify each rating the AI challenged. Record which challenges you accepted and which you rejected, with the reason.

9 Confirm every asset has a named owner. An asset with no owner has no one to accept its risk, which makes the register decorative.

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LAB 4 · VERIFY

Build an Asset Register and Classify What Needs Protecting

✅ Expected result

Every row has an owner, a data classification with a reason, three C/I/A ratings and a criticality. Your register includes at least two non-technical assets that no scanner would have found, and you can name your crown jewels and say what makes each one critical.

IF IT DOESN'T WORK

Everything is rated High

you are rating the technology, not the business impact; force a ranking by asking which single asset you would restore first.

Ownership is unclear

assign the manager of the business function the asset serves, not the system administrator.

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TOPIC 02 · HANDS-ON

AI-Assisted Risk Scoring and the Risk Register

LAB 5

A2 · A4 · K4 — Combine the asset register and the vulnerability worklist into a scored risk register, using AI to sharpen the likelihood estimate.

A risk is not a vulnerability. A risk is a threat exploiting a vulnerability against an asset, with a likelihood and an impact. In this lab you join Lab 3's vulnerabilities to Lab 4's assets to produce a proper risk register, score each entry on a 5x5 likelihood-impact matrix, and use an AI assistant to pressure-test your likelihood estimates against current threat intelligence. You end with a heat map that shows management where the real exposure is.

YOU'LL BUILD

A scored risk register with inherent risk, existing controls and residual risk for every entry, plus a 5x5 heat map showing the distribution.

TOOLCHAIN

Lab 3 worklist, Lab 4 asset register, the Cybersecurity Simulator (alfredang.github.io/cybersecuritysimulator), a spreadsheet, a generative AI assistant

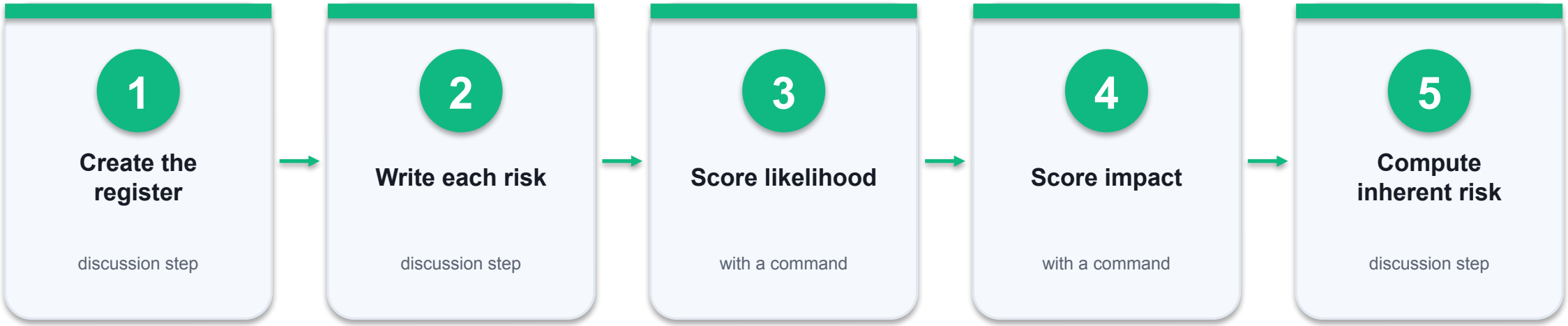
DONE WHEN

Every risk is written as a complete threat-vulnerability-asset-consequence sentence and carries both an inherent and a residual score. Your heat map shows a real spread rather than everything clustered in one cell, and you can justify your top five to a manager.

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LAB 5 · PROCESS MAP

How Lab 5 Runs



THE POINT

A2 · A4 · K4 — Combine the asset register and the vulnerability worklist into a scored risk register, using AI to sharpen the likelihood estimate.

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LAB 5 · PROCEDURE

AI-Assisted Risk Scoring and the Risk Register — Part 1/3

1. Create the risk register with the columns Risk ID, Asset, Threat, Vulnerability, Existing Controls, Likelihood (1-5), Impact (1-5), Inherent Risk, Treatment, Owner, Target Date, Residual Risk.

2. Write each risk as a full sentence before you score it: 'A [threat actor] exploits [vulnerability] on [asset], resulting in [consequence].' A risk you cannot phrase this way is not yet a risk.

3. **Before you score likelihood from a spreadsheet, see the attacks work. Open the Cybersecurity Simulator and run the Phishing, SQL Injection and Social Engineering modules — note for each how little the attacker needs and how plausible the lure is. Likelihood is a judgement about human and technical reality, not a number you invent.**

   → https://alfredang.github.io/cybersecuritysimulator/

4. **Use the simulator's Password Strength Lab to test three passwords of the kind your organisation actually allows. Record the estimated crack time — that figure is the evidence behind any credential-attack likelihood score you write.**

   → https://alfredang.github.io/cybersecuritysimulator/

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LAB 5 · PROCEDURE

AI-Assisted Risk Scoring and the Risk Register — Part 2/3

5 Score Likelihood 1-5 using evidence — is the vulnerability on the CISA KEV list, is the asset internet-facing, is there a public exploit, does an existing control reduce the chance, and what did the simulator show you about how easily this attack lands?

6 Score Impact 1-5 from the asset's criticality in Lab 4, adjusted for what this specific threat would actually do to it.

7 **Compute Inherent Risk as Likelihood x Impact, and band it: 1-4 Low, 5-9 Medium, 10-14 High, 15-25 Critical.**

```
$ Inherent Risk = Likelihood × Impact
```

8 Record the controls already in place for each risk, then re-score to get Residual Risk. The gap between inherent and residual is exactly what your existing controls are buying you.

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LAB 5 · PROCEDURE

AI-Assisted Risk Scoring and the Risk Register — Part 3/3

Ask your AI assistant to challenge your likelihood scores against current threat activity.

9

```
$ Prompt: 'Here is a risk register with likelihood scores from 1 to 5. For each risk, state whether current threat intelligence suggests my likelihood is too high, too low or about right, and give the reason. Note where you are uncertain.'
```

10 Adjust the scores you agree with and document the ones you rejected. A risk register that is never challenged is a risk register nobody believes.

11 Build the 5x5 heat map — likelihood on one axis, impact on the other, each cell counting the risks that land in it — and identify which quadrant carries your exposure.

12 Sort by residual risk and confirm the top five are the risks you would genuinely lose sleep over. If they are not, your scoring is wrong.

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LAB 5 · VERIFY

AI-Assisted Risk Scoring and the Risk Register

✅ Expected result

Every risk is written as a complete threat-vulnerability-asset-consequence sentence and carries both an inherent and a residual score. Your heat map shows a real spread rather than everything clustered in one cell, and you can justify your top five to a manager.

IF IT DOESN'T WORK

Inherent and residual are identical everywhere

you have not recorded the existing controls.

All risks land in the same band

your 1-5 scales are being used as 3-4; force the extremes.

The AI is confidently wrong about a threat

treat its output as a prompt to check the source, never as the source.

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TOPIC 02 · HANDS-ON

Develop the Risk Management Plan

LAB 6

A4 · K4 — Turn the scored register into a risk management plan with treatments, owners, dates and a monitoring cadence that survives contact with reality.

A register records risk; a plan does something about it. In this lab you assign each risk one of the four treatments — mitigate, transfer, avoid or accept — select the specific controls that implement the mitigation, name an owner and a target date for each, and map the whole plan onto the NIST Cybersecurity Framework functions so that the coverage gaps become visible. You then use AI to draft the management summary and you edit it into something you would actually sign.

YOU'LL BUILD

A risk management plan: treatment decisions with named controls, owners, target dates and success criteria; a NIST CSF coverage map; and a one-page management summary.

TOOLCHAIN

Lab 5 risk register, NIST CSF 2.0, a spreadsheet, a generative AI assistant

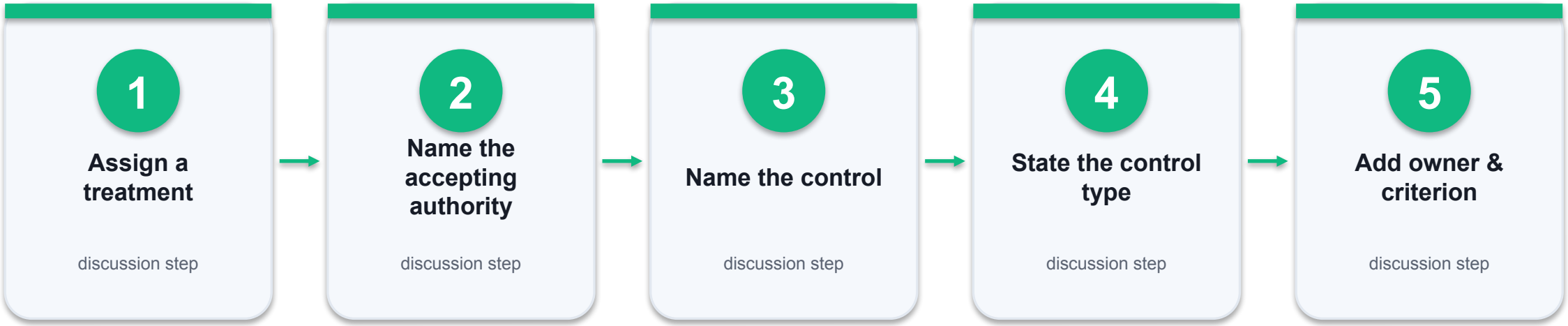
DONE WHEN

Every risk carries a treatment, and every Mitigate carries a named control, an owner, a date and a testable success criterion. Your NIST CSF map has at least one control under all six functions, and the management summary states the top three exposures in business language without jargon.

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LAB 6 · PROCESS MAP

How Lab 6 Runs



THE POINT

A4 · K4 — Turn the scored register into a risk management plan with treatments, owners, dates and a monitoring cadence that survives contact with reality.

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LAB 6 · PROCEDURE

Develop the Risk Management Plan — Part 1/2

1. Assign each risk in the register one of the four treatments — Mitigate, Transfer, Avoid or Accept. Every risk gets exactly one; nothing may be left blank.

2. Every Accept decision needs a named accepting authority and a written rationale. An accepted risk with no name against it is an ignored risk.

3. For each Mitigate decision, name the specific control that will reduce the risk — not 'improve security', but 'disable Telnet on the three switches and enforce SSHv2 with key authentication'.

4. State the control type for each: preventive, detective or corrective. A plan made only of detective controls will find every breach and stop none of them.

5. Add an owner, a target date and a success criterion to every treatment. The success criterion must be testable — 'no Telnet listener on any managed switch, verified by Nmap' is testable; 'switches hardened' is not.

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LAB 6 · PROCEDURE

Develop the Risk Management Plan — Part 2/2

6 **Map every planned control onto the NIST CSF 2.0 functions — Govern, Identify, Protect, Detect, Respond, Recover — and count the controls in each. The thin function is your real gap.**

→ https://www.nist.gov/cyberframework

7 Fill the thinnest function with at least one concrete control. Most first-draft plans are heavy on Protect and empty on Recover.

8 Set the monitoring cadence: which risks are reviewed monthly, which quarterly, and what event forces an out-of-cycle review.

9 **Ask your AI assistant to draft a one-page management summary of the plan for a non-technical audience.**

$ Prompt: 'Summarise this risk management plan for a non-technical executive in under 400 words: the top three exposures in business terms, what we are doing about each, what it needs, and what happens if we do nothing. No jargon.'

10 Edit the draft hard. Cut anything the register does not support, and add the one thing the AI could not know — what this actually costs and who has to approve it.

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LAB 6 · VERIFY

Develop the Risk Management Plan

✅ Expected result

Every risk carries a treatment, and every Mitigate carries a named control, an owner, a date and a testable success criterion. Your NIST CSF map has at least one control under all six functions, and the management summary states the top three exposures in business language without jargon.

IF IT DOESN'T WORK

Everything is Mitigate

some risks are genuinely cheaper to accept or transfer; a plan that mitigates everything will be funded for nothing.

Success criteria are unverifiable

rewrite each one as a test someone could run.

Recover is empty

add backup restoration testing.

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TOPIC RECAP

Recap — Topic 02

1

Lab 4 — Build an Asset Register and Classify What Needs Protecting

Determine which organisational assets require protection, and defend the classification with a business impact judgement rather than a gut feeling.

2

Lab 5 — AI-Assisted Risk Scoring and the Risk Register

Combine the asset register and the vulnerability worklist into a scored risk register, using AI to sharpen the likelihood estimate.

3

Lab 6 — Develop the Risk Management Plan

Turn the scored register into a risk management plan with treatments, owners, dates and a monitoring cadence that survives contact with reality.

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03

TOPIC 03

Designing and Implementing AI-Enhanced Network Security Measures

Defence in depth · planning/building/management phases · segmentation & zero trust · AI firewalls, IDS/IPS, NAC · AI security itself

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TOPIC 03 · ROADMAP

Topic 03 — What It Covers and How It Is Assessed

1

Learning outcome

LO3 — Design and implement security measures for network components by managing security phases.

2

Knowledge assessed (WA-SAQ)

K3: Types and techniques of network security measures

3

Abilities assessed (Case Study)

A5: Define planning, building and management phases for network security design · A6: Develop security measures for network components · A7: Design auditing and incident response procedures

4

Hands-on labs

Labs 7, 8, 9 — 30% of course time.

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TOPIC WEIGHTING 30%

Key Concepts — Topic 03

1. Defence in depth layers physical, technical and administrative controls so that the failure of any single control does not become a breach.

2. Network security design runs in three phases — planning (requirements, risk, architecture), building (configure, harden, test) and management (monitor, tune, review).

3. Segmentation and zero trust shrink the blast radius: VLANs, DMZs and micro-segmentation stop lateral movement, and 'never trust, always verify' removes implicit network trust.

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TOPIC WEIGHTING 30%

Key Concepts — Topic 03 (continued)

1. AI-powered firewalls and NGFW policy engines adapt rule sets to observed traffic instead of relying purely on static, hand-written rules.

2. AI-enhanced IDS/IPS, UEBA and adaptive access control score every session on device, behaviour, location and time, and step up authentication when risk rises.

3. The AI you deploy is itself an asset to defend — NIST AI 100-2 catalogues evasion, poisoning, privacy and misuse attacks, and the NSA/CISA guidance tells you how to deploy AI securely.

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Defence in Depth — the Three Control Categories

1

Physical controls

Locks, badge access, CCTV, secure racks and cable management. If someone can touch the switch, the rest is negotiable.

2

Technical controls

Firewalls, IDS/IPS, encryption, authentication, endpoint protection, segmentation — the controls this course configures.

3

Administrative controls

Policies, standards, procedures, training and review. These are what keep the technical controls correct over time.

4

Preventive

Stops the event happening — a firewall rule, an access control, patching.

5

Detective

Notices that it happened — logging, IDS, anomaly detection, audit.

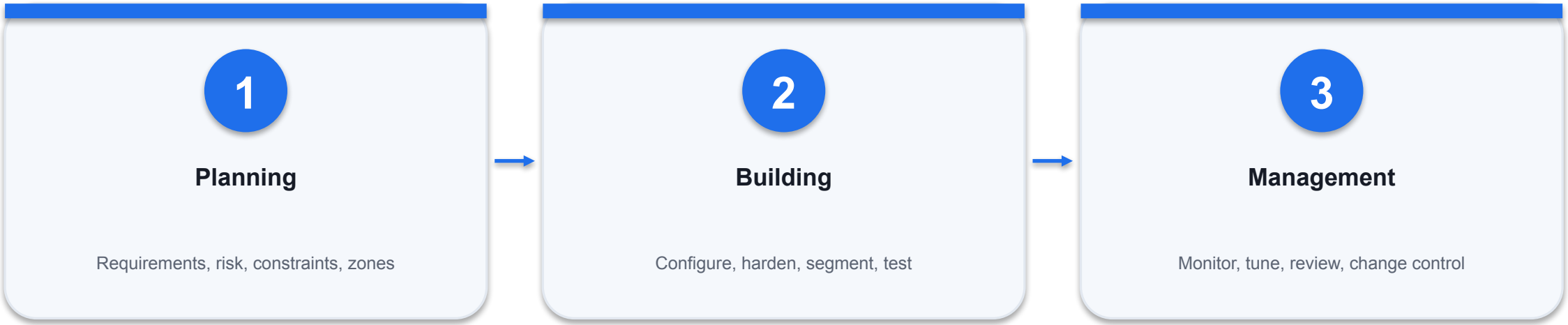
6

Corrective

Restores after it happened — backups, failover, the incident response playbook.

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The Three Phases of Network Security Design



A5 IN ONE LINE

Define what the design must achieve (planning), make it real and prove it works (building), and keep it correct as the network changes (management). A design with no management phase decays within a quarter.

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Segmentation and Zero Trust

1

Why segment

A flat network means one compromised laptop can reach the domain controller. Segmentation makes lateral movement expensive.

2

VLANs and subnets

The basic separation — different departments, different functions, different trust levels.

3

DMZ

A buffer zone for internet-facing services, so a compromised web server is not inside the internal network.

4

Micro-segmentation

Per-workload policy rather than per-subnet — the granular end of the same idea.

5

Zero trust

Never trust, always verify. Location on the network stops being a reason to grant access.

6

Default deny

The rule that makes segmentation real: allow what is explicitly required, block everything else.

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Access Control, Authentication and Secure Remote Access

1

Least privilege

Every account and service gets the minimum access it needs to do its job — and no more. The single most effective limit on what a compromise can reach.

2

Access control models

Discretionary (DAC), Role-Based (RBAC) and Attribute-Based (ABAC). RBAC is the practical default: permissions attach to a role, not to a person.

3

ACLs

Access Control Lists on routers and switches enforce which source may reach which destination on which port — the device-level expression of the zone matrix.

4

AAA

Authentication, Authorisation and Accounting via RADIUS or TACACS+ — centralises who may log in, what they may do, and what they did.

5

Multi-factor authentication (MFA)

Something you know plus something you have or are. It defeats credential stuffing and password reuse, which is how most account takeover starts.

6

VPN for remote access

An encrypted tunnel over an untrusted network. Terminate it into a restricted zone with individual accounts and MFA — never a shared account into the flat internal network.

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AI-Enhanced Controls You Will Specify

1

AI-powered firewall

Adapts rule sets to observed traffic patterns instead of relying purely on static hand-written rules.

2

AI-enhanced IDS/IPS

Learns normal protocol and volume behaviour per segment, so it flags what a static threshold would miss.

3

UEBA

User and Entity Behaviour Analytics — profiles each user and device against its own history, not a global norm.

4

Adaptive access control

Scores device posture, behaviour, location and time, then steps up authentication or blocks as risk rises.

5

NAC with AI classification

Fingerprints and classifies devices as they join, so an unknown device does not silently get a trusted VLAN.

6

AI policy optimisation

Recommends rule changes from observed traffic, and finds the shadowed and over-broad rules humans leave behind.

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NIST AI 100-2 — Attacks Against AI Systems

Attack class	What the adversary does	Against your NDR model	Primary mitigation
Evasion	Crafts input that the deployed model misclassifies	Slows a scan until it falls inside the baseline	Adversarial training; ensemble and rate-independent features
Poisoning	Corrupts the training data or the model itself	Seeds 'normal-looking' attack traffic during learning	Data provenance, integrity verification, anomaly checks on training data
Privacy	Extracts training data or model details from outputs	Infers your network topology from detection behaviour	Output minimisation, differential privacy, access control
Misuse (GenAI)	Abuses a generative model to produce harmful output	Turns your security copilot into a recon assistant	Guardrails, prompt filtering, least-privilege tool access

WHEN IT MATTERS NIST AI 100-2e2025 covers predictive AI (evasion, poisoning, privacy) and adds misuse for generative AI. MITRE ATLAS maps the same ground as adversary tactics and techniques.

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NSA / CISA — Deploying AI Systems Securely

1

Secure the environment

Governance, architecture, hardened config

2

Protect the AI system

Validate, encrypt, control access, monitor

3

Secure operation & maintenance

Audit, update, respond, retire safely

THE THREE CATEGORIES

The April 2024 joint Cybersecurity Information Sheet from NSA, CISA, FBI and international partners organises AI deployment security into these three stages. Lab 9 turns them into your hardening checklist.

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Protecting the Data Your AI Learns From

1

Provenance tracking

Record where every dataset came from — cryptographically signed, append-only ledgers where the stakes justify it.

2

Integrity verification

Cryptographic hashes and checksums on data at rest and in transit, so silent corruption is detectable.

3

Digital signatures

Sign dataset revisions so an unauthorised change to training data cannot pass unnoticed.

4

Classification & access control

Apply sensitivity-based access to training data, exactly as you would to production data.

5

Encryption

Strong encryption in transit and at rest across every stage of the pipeline, not only the final store.

6

Continuous assessment

Re-assess data risk on a schedule — data drift degrades a model quietly, without any alert firing.

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WHAT YOU'LL DO

Hands-On Labs — Topic 03

Lab 7

- Design the Layered Architecture Across the Three Security Phases
-
- Define the planning, building and management phases for a network security design, and express the design as layered def

Lab 8

- Implement AI-Enhanced Controls — Firewall Policy, IDS and Adaptive Access
-
- Develop the security measures for the network components: build a default-deny firewall policy, tune detection rules, an

Lab 9

- Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls
-
- Recognise that the AI you deploy is an attack surface, and apply the recognised controls for deploying an AI system secu

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TOPIC 03 · HANDS-ON

Design the Layered Architecture Across the Three Security Phases

LAB 7

A5 · K3 — Define the planning, building and management phases for a network security design, and express the design as layered defence-in-depth controls.

Security design is a phased engineering activity, not a shopping list. In this lab you take the risk management plan from Lab 6 and produce an actual design: you write the requirements and constraints in the planning phase, draw a segmented target architecture with the controls placed at each layer, and define the management phase that keeps the design honest after go-live. You use an AI assistant as a design reviewer and then defend the design against its objections.

YOU'LL BUILD

A three-phase security design: a planning-phase requirements table, a segmented target architecture diagram with controls at every layer, and a management-phase operating model.

TOOLCHAIN

Lab 6 plan, the IP Calculator (alfredang.github.io/ipcalculator), a diagramming tool (draw.io or paper), a generative AI assistant

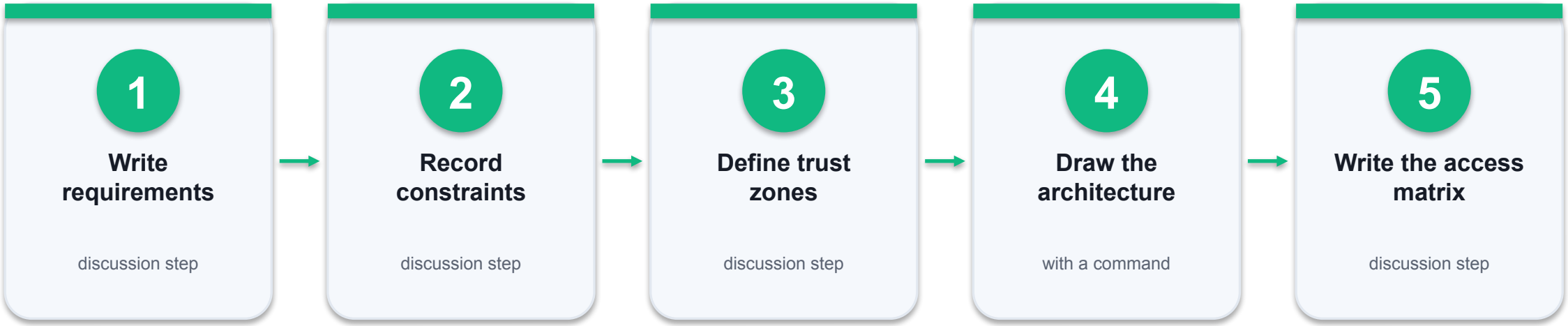
DONE WHEN

Every security requirement traces to a risk ID, the access matrix is default-deny with justified exceptions, all three control categories are present, and every High or Critical risk maps to a control or a documented acceptance. You can answer all three of the AI's attack paths.

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LAB 7 · PROCESS MAP

How Lab 7 Runs



THE POINT

A5 · K3 — Define the planning, building and management phases for a network security design, and express the design as layered defence-in-depth controls.

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LAB 7 · PROCEDURE

Design the Layered Architecture Across the Three Security Phases — Part 1/3

1. Planning phase — write the security requirements the design must satisfy, each traced back to a specific risk ID from Lab 5. A requirement with no risk behind it is a preference.

2. Planning phase — record the constraints: budget, existing equipment, skills available, downtime tolerance and any regulatory obligation. Constraints are what make a design real.

3. Planning phase — define the trust zones: Internet, DMZ, internal user network, server network, management network and any OT or guest segment. State what each zone is allowed to reach.

4. **Planning phase — turn those zones into real address space. Use the IP Calculator's recursive subnet splitting to carve your allocation into one subnet per zone, sized to the hosts each actually needs. Record the CIDR, usable range and host count per zone — a segmentation design with no addressing plan cannot be implemented.**

→ https://alfredang.github.io/ipcalculator/

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LAB 7 · PROCEDURE

Design the Layered Architecture Across the Three Security Phases — Part 2/3

5 Building phase — draw the target architecture with the zones as segments and place the controls at each layer: perimeter firewall, DMZ, internal segmentation, host controls and data controls.

6 Building phase — write the zone-to-zone access matrix. Rows are source zones, columns are destination zones, cells hold the permitted protocols. Default deny: every empty cell is blocked.

7 Building phase — classify every control you have placed as physical, technical or administrative, and confirm you have all three. A design with no administrative controls has no way to stay correct.

8 Management phase — define what happens after go-live: who monitors what, the review cadence, the change-control path, and the trigger that forces a redesign.

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LAB 7 · PROCEDURE

Design the Layered Architecture Across the Three Security Phases — Part 3/3

Ask your AI assistant to review the design as an attacker would.

9

```
$ Prompt: 'Here is a segmented network security design with a zone access matrix. Acting as an attacker, describe three paths you would take through this architecture and name the control that would have to fail at each step. Then identify the weakest zone boundary.'
```

10 Answer each attack path in writing: either add a control that closes it, or record why the residual risk is acceptable and who accepts it.

11 Trace every High and Critical risk from Lab 5 to at least one control in the design. Any risk with no control is either untreated or was accepted — and it must say which.

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LAB 7 · VERIFY

Design the Layered Architecture Across the Three Security Phases

✅ Expected result

Every security requirement traces to a risk ID, the access matrix is default-deny with justified exceptions, all three control categories are present, and every High or Critical risk maps to a control or a documented acceptance. You can answer all three of the AI's attack paths.

IF IT DOESN'T WORK

The matrix has more allow cells than deny

you are documenting today's flat network rather than designing a segmented one.

No management phase

a design with no operating model degrades within a quarter.

The AI's attack paths seem unrealistic

say why in writing; that reasoning is the deliverable.

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TOPIC 03 · HANDS-ON

Implement AI-Enhanced Controls — Firewall Policy, IDS and Adaptive Access

LAB 8

A6 · K3 — Develop the security measures for the network components: build a default-deny firewall policy, tune detection rules, and specify adaptive AI-driven access control.

This is where the design becomes configuration. You write a default-deny firewall ruleset from the Lab 7 access matrix, run detection against real traffic to see what your rules actually catch, and specify the risk-based conditions under which an AI-driven access control should step up authentication or block outright. You then compare a static rule against a behavioural model on the same traffic — and see exactly where each one wins.

YOU'LL BUILD

A default-deny firewall ruleset derived from the access matrix, a tuned detection ruleset with its true/false positive counts, and an adaptive access-control policy specification.

TOOLCHAIN

Firewall rule syntax (iptables or a NGFW policy table), the Cryptography Toolkit (alfredang.github.io/cryptography-toolkit), Wireshark, Python, a generative AI assistant

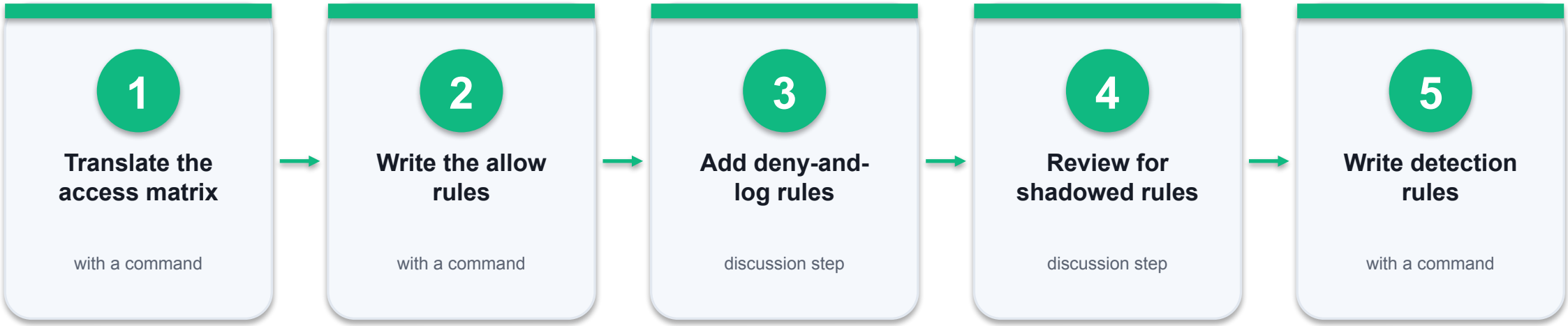
DONE WHEN

Your ruleset ends in an explicit default deny, every allow rule carries a business-reason comment traced to a requirement, and you have no shadowed rules. You can state the true/false positive count for each detection rule and say which of the static rule or the AI model you would deploy for each of the three detections, and why.

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LAB 8 · PROCESS MAP

How Lab 8 Runs



THE POINT

A6 · K3 — Develop the security measures for the network components: build a default-deny firewall policy, tune detection rules, and specify adaptive AI-driven access control.

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LAB 8 · PROCEDURE

Implement AI-Enhanced Controls — Firewall Policy, IDS and Adaptive Access — Part 1/3

1. **Translate the Lab 7 access matrix into an ordered firewall ruleset. Specific allow rules first, the default deny last — order is the whole game in a firewall.**

```
$ # Rule 1: allow established return traffic
```

2. **Write each allow rule with source zone, destination zone, protocol, port, and a comment naming the business reason. An uncommented rule becomes permanent because nobody dares delete it.**

```
$ iptables -A FORWARD -s 10.0.10.0/24 -d 10.0.30.0/24 -p tcp --dport 443 -j ACCEPT -m comment --comment 'Users to app servers HTTPS - REQ-004'
```

3. Add explicit deny-and-log rules for the traffic you most want to know about — management protocols crossing a zone boundary, and any outbound connection from the server zone.

4. Review the ruleset for shadowed rules: any rule that can never match because an earlier rule already covers its traffic. Shadowed rules are the most common firewall defect there is.

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LAB 8 · PROCEDURE

Implement AI-Enhanced Controls — Firewall Policy, IDS and Adaptive Access — Part 2/3

5 **Your ruleset controls WHO may connect; encryption controls what an interceptor can read. Use the Cryptography Toolkit to encrypt a sample management credential with AES-256 (CBC), then decrypt it. Then try the same string with DES and RC4 — both are flagged insecure for a reason. Record why your design mandates AES for management traffic.**

```
→ https://alfredang.github.io/cryptography-toolkit/
```

6 Now the detection layer. Write three detection rules against the Lab 2 capture — one for a port scan, one for a large outbound transfer, and one for a connection to a management port from a user subnet.

7 **Run each rule over the capture and count the true positives and false positives. A rule with more false positives than true is worse than no rule, because it trains the analyst to ignore alerts.**

```
$ tshark -r traffic.pcapng -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields -e ip.src | sort | uniq -c | sort -rn
```

8 Compare the static rule against the Lab 2 anomaly model on the same traffic. Record where the rule wins (precision on known patterns) and where the model wins (unnamed behaviour).

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LAB 8 · PROCEDURE

Implement AI-Enhanced Controls — Firewall Policy, IDS and Adaptive Access — Part 3/3

9 Specify the adaptive access-control policy: the signals (device posture, location, time, behaviour deviation, resource sensitivity), the risk bands, and the action at each band — allow, step-up MFA, restrict, or block.

10 State the human-in-the-loop boundary explicitly. Which actions may the system take autonomously, and which require a named person to approve? Automatic blocking of a production server needs an owner.

11 **Ask your AI assistant to review the firewall ruleset for shadowed rules, overly broad allows and missing egress controls, then verify each finding by reading the rule yourself.**

```
$ Prompt: 'Review this ordered firewall ruleset. Identify shadowed rules that can never match, allow rules that are broader than their stated business reason, and any missing egress control. Quote the rule number for each finding.'
```


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LAB 8 · VERIFY

Implement AI-Enhanced Controls — Firewall Policy, IDS and Adaptive Access

✅ Expected result

Your ruleset ends in an explicit default deny, every allow rule carries a business-reason comment traced to a requirement, and you have no shadowed rules. You can state the true/false positive count for each detection rule and say which of the static rule or the AI model you would deploy for each of the three detections, and why.

IF IT DOESN'T WORK

The default deny is not last

everything after it is dead code.

All traffic is blocked

you are missing the ESTABLISHED,RELATED rule, so return traffic never gets back.

The detection rule fires constantly

your threshold is tuned to a quiet lab, not to production; raise it against the baseline.

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TOPIC 03 · HANDS-ON

Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls

LAB 9

A5 · A6 · A7 · K3 — Recognise that the AI you deploy is an attack surface, and apply the recognised controls for deploying an AI system securely.

Every previous lab used AI as a defensive tool. This one turns the question around: what happens when the AI is the target? You run an evasion attack against your own Lab 2 anomaly model and watch it go blind, simulate a poisoning attack on its training data, then work through the NIST AI 100-2 attack taxonomy and the NSA/CISA 'Deploying AI Systems Securely' controls to produce a hardening checklist for the AI components in your Lab 7 design.

YOU'LL BUILD

A demonstrated evasion and a demonstrated poisoning attack against your own model, plus an AI security hardening checklist mapped to the NSA/CISA three categories and the NIST AI 100-2 attack classes.

TOOLCHAIN

Lab 2 model, Python, scikit-learn, NIST AI 100-2e2025, NSA/CISA CSI, MITRE ATLAS

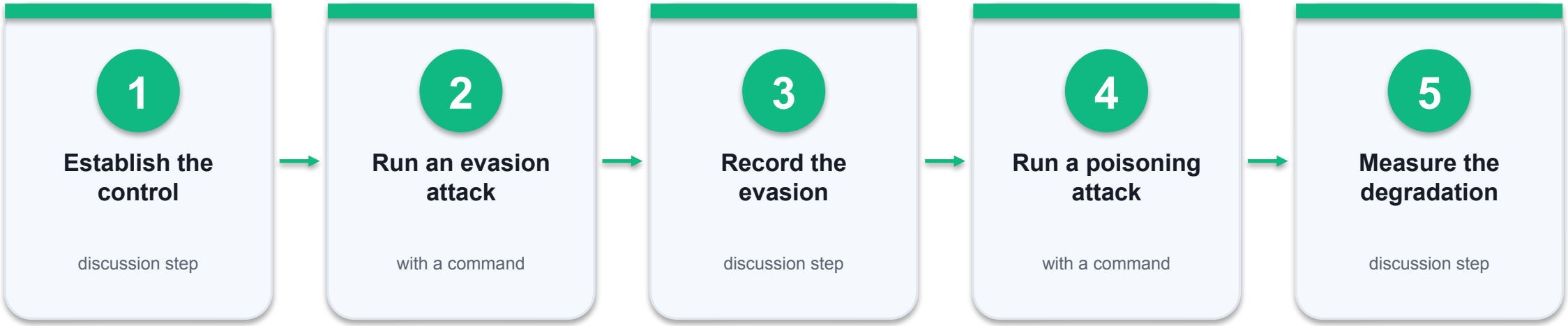
DONE WHEN

You can demonstrate the slowed scan evading a model that detected the fast one, and you can state the measured detection drop after poisoning. Your checklist covers all three NSA/CISA categories, names an owner and a verification method for every control, and includes a stated fallback for when the AI is down.

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LAB 9 · PROCESS MAP

How Lab 9 Runs



THE POINT

A5 · A6 · A7 · K3 — Recognise that the AI you deploy is an attack surface, and apply the recognised controls for deploying an AI system securely.

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LAB 9 · PROCEDURE

Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls — Part 1/3

1. Reload your Lab 2 Isolation Forest and confirm it still detects the port scan. This is your control condition — you cannot show an attack worked without it.

2. **Evasion attack: slow the scan down. Regenerate the scan features with a long inter-packet delay so the connection rate falls inside the baseline, then re-score it.**

```
$ nmap -sS -T1 --scan-delay 5s -p 1-1024 192.168.56.101
```

3. Record what happened. The traffic is functionally identical, the intent is identical, and the model no longer flags it — that is an evasion attack, and it needed no access to the model at all.

4. **Poisoning attack: inject slow-scan traffic into the training window, refit the model, and re-score the original fast scan. Measure how far the detection degrades.**

```
$ X_poisoned = np.vstack([X_baseline, X_slowscan])
```


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LAB 9 · PROCEDURE

Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls — Part 2/3

5 Record the degradation as a number — detection rate before poisoning versus after. 'The model got worse' is an anecdote; a measured drop is evidence.

6 **Map both attacks onto the NIST AI 100-2e2025 taxonomy — evasion, poisoning, privacy and misuse — and state which class each belongs to and at what stage of the ML lifecycle it lands.**

→ https://csrc.nist.gov/pubs/ai/100/2/e2025/final

7 **Look up the matching tactics and techniques in MITRE ATLAS, the adversarial threat knowledge base for AI systems, and note the mitigations it lists.**

→ https://atlas.mitre.org/

8 **Work through the NSA/CISA 'Deploying AI Systems Securely' controls under its three categories — secure the deployment environment, continuously protect the AI system, and secure AI operation and maintenance.**

→ https://www.cisa.gov/news-events/alerts/2024/04/15/joint-guidance-deploying-ai-systems-securely

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LAB 9 · PROCEDURE

Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls — Part 3/3

9 Add the CISA/NSA AI data-security controls that protect the training pipeline: data provenance tracking, cryptographic integrity verification, digital signatures on dataset revisions, access control by classification, encryption in transit and at rest, and continuous risk assessment.

10 Produce the hardening checklist for the AI components in your Lab 7 design. Every item names the control, the attack class it addresses, who owns it and how it is verified.

11 Add the one control that no framework will hand you: define what the security operation does when the AI is unavailable or untrusted. A detection capability with no manual fallback is a single point of failure.

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LAB 9 · VERIFY

Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls

✅ Expected result

You can demonstrate the slowed scan evading a model that detected the fast one, and you can state the measured detection drop after poisoning. Your checklist covers all three NSA/CISA categories, names an owner and a verification method for every control, and includes a stated fallback for when the AI is down.

IF IT DOESN'T WORK

The slow scan is still detected

your feature set does not include a rate term, so slowing it changes nothing; add connections-per-second and retry.

Poisoning shows no effect

your injected sample is too small relative to the baseline; raise the proportion and re-measure.

Results vary between runs

set random_state so the comparison is fair.

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TOPIC RECAP

Recap — Topic 03

1

Lab 7 — Design the Layered Architecture Across the Three Security Phases

Define the planning, building and management phases for a network security design, and express the design as layered defence-in-depth controls.

2

Lab 8 — Implement AI-Enhanced Controls — Firewall Policy, IDS and Adaptive Access

Develop the security measures for the network components: build a default-deny firewall policy, tune detection rules, and specify adaptive AI-driven access control.

3

Lab 9 — Secure the AI Itself — Adversarial ML and the NSA/CISA Deployment Controls

Recognise that the AI you deploy is an attack surface, and apply the recognised controls for deploying an AI system securely.

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04

TOPIC 04

AI-Assisted Security Incident Documentation and Response

Logging & syslog · SIEM/SOAR · AI alert triage · the NIST IR lifecycle · incident documentation · playbooks & post-incident review

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Topic 04 — What It Covers and How It Is Assessed

1

Learning outcome

LO4 — Document security incidents and apply incident response configurations to manage risks based on established procedures.

2

Knowledge assessed (WA-SAQ)

K4: Network security implementation risk management plans and procedures

3

Abilities assessed (Case Study)

A8: Document security incidents · A9: Implement configurations aligned with incident response procedure design

4

Hands-on labs

Labs 10, 11, 12 — 20% of course time.

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TOPIC WEIGHTING 20%

Key Concepts — Topic 04

1. Logging is the evidence base of incident response — syslog, Windows Event Log, flow records and firewall logs must be centralised, time-synchronised and retained.

2. SIEM correlates events into incidents; SOAR executes the response playbook — isolate the host, revoke the credential, block the address — at machine speed.

3. AI alert triage attacks the real bottleneck: it deduplicates, clusters and contextualises so the analyst sees a handful of ranked incidents instead of thousands of raw alerts.

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TOPIC WEIGHTING 20%

Key Concepts — Topic 04 (continued)

1. The NIST SP 800-61 lifecycle is Preparation, Detection & Analysis, Containment/Eradication/Recovery, and Post-Incident Activity — AI accelerates each step but owns none of them.

2. An incident record must state what happened, when, to which assets, the severity, the actions taken, the evidence, and the person accountable — this is the A8 deliverable.

3. Human-in-the-loop approval remains mandatory for destructive or wide-blast-radius actions; AI recommends, a named responder decides.

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Logging — the Evidence Base of Incident Response

1

Syslog

The standard for network devices and Unix hosts; forwards to a central collector over UDP/TCP 514 or TLS.

2

Windows Event Log

Security, System and Application channels; forwarded with WEF or an agent to the same collector.

3

Flow records

NetFlow, sFlow and IPFIX summarise who talked to whom, when and how much — cheap, and enough for most correlation.

4

Firewall & IDS logs

Allow and deny decisions plus signature hits — the boundary record of what tried to cross.

5

Time synchronisation

Every source on NTP against the same reference. Skewed clocks make cross-source correlation worthless.

6

Retention & integrity

Keep long enough to investigate a slow intrusion, and store so that an attacker with local admin cannot rewrite it.

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SIEM vs SOAR vs XDR

	SIEM	SOAR	XDR
Primary job	Collect and correlate events into incidents	Execute the response playbook	Detect and respond across layers in one platform
Core input	Logs from everywhere	An alert plus a playbook	Endpoint, network, identity and cloud telemetry
Core output	A prioritised incident	An action taken automatically	A correlated incident with response built in
Where AI helps	Correlation, deduplication, ranking	Deciding which playbook and when	Cross-layer correlation and root cause
Human role	Investigate and decide	Approve high-impact actions	Investigate, and own the decision

WHEN IT MATTERS The names blur between vendors. Judge a platform by what it actually does with your telemetry, not by which of these three words appears on the datasheet.

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LO4 · SOLVING THE REAL BOTTLENECK

AI-Assisted Alert Triage



WHY THIS MATTERS MOST

Detection is rarely the failure point — the failure is that the true alert was sitting in a queue nobody could read. Triage is where AI returns the most analyst time.

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LO4 · K4 · THE LIFECYCLE

The NIST SP 800-61 Incident Response Lifecycle



AI ACCELERATES, IT DOES NOT OWN

AI speeds up detection, scoping and drafting. Classification, containment authority and the lessons-learned judgement stay with a named responder.

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What a Defensible Incident Record Contains

1. **Identification**
   Incident ID, detection time, who detected it, who is handling it, current status.

2. **Evidence-based timeline**
   Every entry timestamped in UTC with the log line or capture that proves it.

3. **Classification**
   Severity, with the written criteria that produced it — so the rating can be reviewed, not just asserted.

4. **Affected assets**
   Named from the asset register, with criticality and the data classification each one holds.

5. **Actions taken**
   Each with a timestamp and the person who took it, including the options considered and rejected.

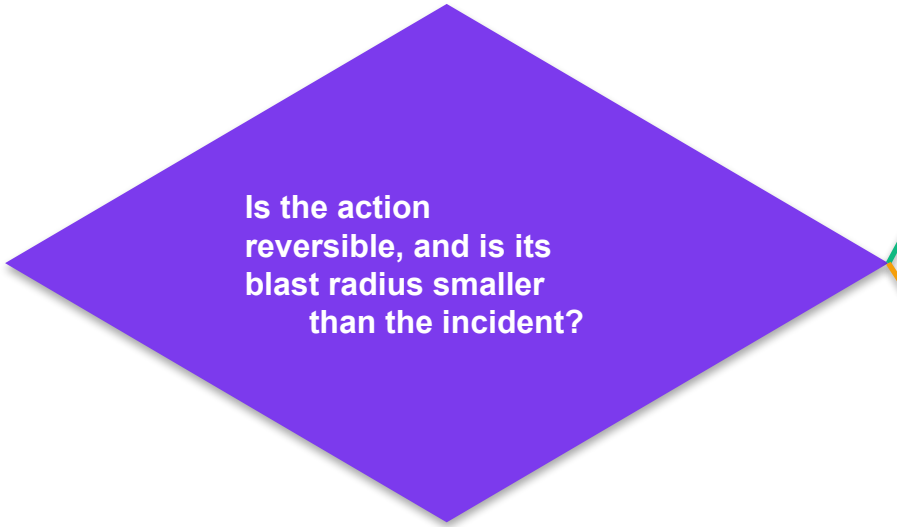
6. **Evidence inventory**
   Every artefact with its storage location, its hash and who has access to it.

7. **Notifications**
   Who was told, when, and which obligations were considered — including PDPA notification to the PDPC where engaged.

8. **Lessons learned**
   What detected it, what delayed the response, and the one control change that goes back into the plan.

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Can This Response Step Be Automated?



YES — automate it

Block a single source address, quarantine one file, expire one session, raise an alert. Fast, reversible, contained.

NO — require approval

Isolate a production server, disable an executive's account, block an entire subnet. A named person approves.

Ask the question before the incident, not during it. An automated action that takes a hospital's patient system offline to stop a low-severity alert is itself the incident.

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WHAT YOU'LL DO

Hands-On Labs — Topic 04

Lab 10

- Centralise Logs and Triage Alerts with AI
-
- Design the auditing arrangement that makes incident response possible, then use AI to reduce a flood of raw alerts into

Lab 11

- Document a Security Incident to the Standard
-
- Produce an incident record that is complete enough to hand to a manager, an auditor or a regulator without a conversatio

Lab 12

- Implement the Incident Response Configuration and Playbook
-
- Implement configurations aligned to the incident response design: containment controls, an automated playbook and the hu

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TOPIC 04 · HANDS-ON

Centralise Logs and Triage Alerts with AI

LAB 10

A7 · K4 — Design the auditing arrangement that makes incident response possible, then use AI to reduce a flood of raw alerts into a ranked, workable queue.

Incident response is only as good as the evidence available when it starts. In this lab you define what must be logged and for how long, work with a realistic multi-source log set, and then attack the actual bottleneck of a security operation: alert volume. You deduplicate, cluster and enrich the alerts, use an AI assistant to rank them and explain its reasoning, and then measure whether the ranking was any good.

YOU'LL BUILD

A logging and audit specification, plus a triaged alert queue ranked by priority with an AI-generated rationale for each and your own verification of the top ten.

TOOLCHAIN

syslog, Windows Event Log, RegexLab (alfredang.github.io/regexgenerator), Python, pandas, a generative AI assistant

DONE WHEN

Your logging specification names the source, events, retention and central destination for every asset class. Deduplication and clustering reduce the queue by a measurable percentage that you can state, and you can report the triage precision of the top ten with the evidence that decided each one.

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LAB 10 · PROCESS MAP

How Lab 10 Runs



THE POINT

A7 · K4 — Design the auditing arrangement that makes incident response possible, then use AI to reduce a flood of raw alerts into a ranked, workable queue.

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LAB 10 · PROCEDURE

Centralise Logs and Triage Alerts with AI — Part 1/3

1. Write the logging specification: for each asset class name the log source, the events that must be captured, the retention period and where the log is centralised.

2. Include the events that matter for investigation — authentication success and failure, privilege escalation, configuration change, firewall deny, and process or service creation. A log of only failures cannot reconstruct an intrusion.

3. **Specify time synchronisation. Every source must use NTP against the same reference, because a correlation across sources with skewed clocks is worthless.**

```
$ timedatectl set-ntp true && timedatectl status
```

4. Confirm the log integrity controls: append-only or write-once storage, restricted access, and forwarding off the host — an attacker with local admin will clear the local log first.

5. **Load the provided multi-source alert set into pandas and look at the raw volume. This is the problem the rest of the lab solves.**

```
$ python3 triage.py   # provided in labs/lab-10/
```


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LAB 10 · PROCEDURE

Centralise Logs and Triage Alerts with AI — Part 2/3

6 **Before filtering at scale, build the pattern and prove it works. Paste a representative log line into RegexLab, write a regex that extracts the source IP and the event type, and check the live match count and the match explanation. A filter you have not tested against real lines is how a triage pipeline silently drops evidence.**

```
→ https://alfredang.github.io/regexgenerator/
```

7 **Use the same tool to write a pattern that catches authentication FAILURES but not successes, then deliberately test it against a success line to confirm it does not match. Record both results — the negative test is the one that matters.**

```
→ https://alfredang.github.io/regexgenerator/
```

8 **Deduplicate: collapse identical alerts on the same source, destination and signature within a time window into one alert with a count. Record how much this alone reduces the queue.**

```
$ df.groupby(['src','dst','signature']).agg(count=('ts','size'), first=('ts','min'), last=('ts','max'))
```

9 Cluster the remaining alerts by source and by time so that a burst from one host reads as one incident rather than forty unrelated events.

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LAB 10 · PROCEDURE

Centralise Logs and Triage Alerts with AI — Part 3/3

10 Enrich each cluster with context from your earlier labs — the asset criticality from Lab 4 and the risk score from Lab 5. Context is what turns an alert into a priority.

11 **Ask your AI assistant to rank the clusters and explain each ranking.**

```
$ Prompt: 'Here are deduplicated, enriched security alert clusters with asset criticality. Rank them by investigation priority. For each, state the likely attack technique, the evidence supporting it, and what a responder should check first. Flag any cluster where the evidence is ambiguous.'
```

12 Verify the top ten yourself against the raw logs. Count how many were genuinely high priority — that number is your triage precision, and it is what you report.

13 Write down the two alert types the AI ranked poorly and say what context it was missing. That gap is the enrichment you add next time.

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LAB 10 · VERIFY

Centralise Logs and Triage Alerts with AI

✅ Expected result

Your logging specification names the source, events, retention and central destination for every asset class. Deduplication and clustering reduce the queue by a measurable percentage that you can state, and you can report the triage precision of the top ten with the evidence that decided each one.

IF IT DOESN'T WORK

Deduplication removes real incidents

your window is too wide; a repeated login failure over hours is a different event from forty in ten seconds.

Everything ranks as critical

the asset-criticality join failed and every alert inherited the default.

Timestamps do not line up

sources are in different time zones; normalise to UTC before correlating.

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TOPIC 04 · HANDS-ON

Document a Security Incident to the Standard

LAB 11

A8 · K4 — Produce an incident record that is complete enough to hand to a manager, an auditor or a regulator without a conversation attached.

An incident that is badly documented is an incident that teaches nothing and proves nothing. Working from the top-priority cluster in Lab 10, you reconstruct the timeline from the raw evidence, classify the severity against defined criteria, and write the full incident record: what happened, when, to which assets, what was done, what the evidence is and who was accountable. You use AI to draft the narrative and then you correct it against the log — because the draft will confidently state things the evidence does not.

YOU'LL BUILD

A complete incident record with a reconstructed evidence-based timeline, a justified severity classification, the actions taken, the evidence inventory and the notification decisions.

TOOLCHAIN

Lab 10 alert queue and raw logs, the PCAP Analyzer and Cryptography Toolkit, NIST SP 800-61, an incident record template, a generative AI assistant

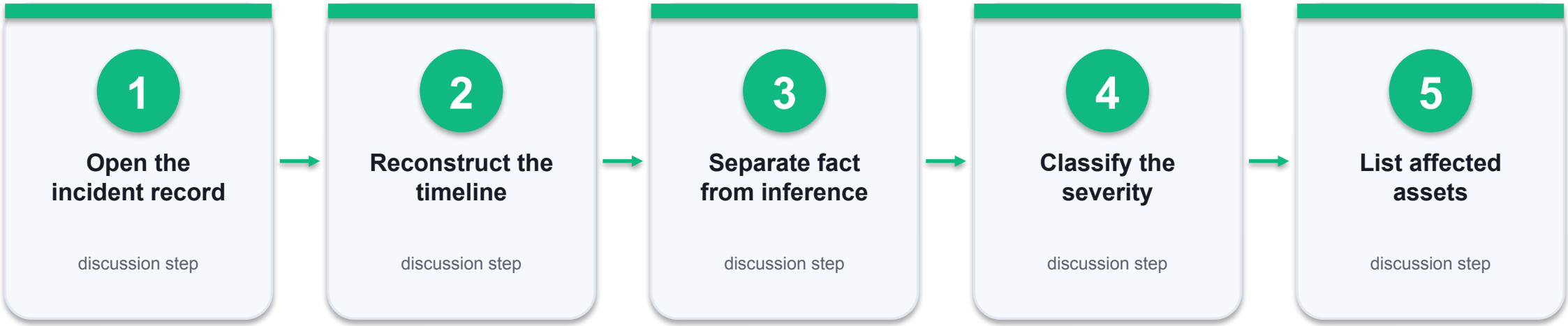
DONE WHEN

Every timeline entry cites its evidence source, inferences are marked as inferences, the severity classification states the criteria used, and the evidence inventory carries a hash for every item. You can name at least one unsupported statement you removed from the AI draft.

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LAB 11 · PROCESS MAP

How Lab 11 Runs



THE POINT

A8 · K4 — Produce an incident record that is complete enough to hand to a manager, an auditor or a regulator without a conversation attached.

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LAB 11 · PROCEDURE

Document a Security Incident to the Standard — Part 1/3

1. Open an incident record and fill the header: incident ID, date and time of detection, detected by, reporter, current status and the assigned handler.
2. Reconstruct the timeline from the raw logs in UTC. Every entry needs a timestamp, the event, the source system and the log line that proves it. A timeline entry with no evidence reference is a claim, not a finding.
3. Separate what you know from what you infer. Write inferences as inferences — 'the account was used from an unrecognised address, which suggests credential compromise' is honest; stating the compromise as fact is not.
4. Classify the severity against written criteria: assets affected, data classification involved, business impact and whether the activity is ongoing. State the criteria you applied so the rating can be reviewed.
5. List the affected assets by name from the Lab 4 register, with their criticality and the data classification each one holds.

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LAB 11 · PROCEDURE

Document a Security Incident to the Standard — Part 2/3

6 Record the response actions taken with a timestamp and the name of the person who took each one. Include the actions that were considered and rejected, and why — a reviewer will ask.

7 **If a packet capture is part of your evidence, summarise it before you cite it. Open it in the PCAP Analyzer and record the capture window, the packet and byte counts, and the conversations involving the affected asset — that summary is what a reader who cannot open a pcap actually needs.**

```
→ https://alfredang.github.io/pcapanalyzer/
```

8 **Build the evidence inventory: log extracts, packet captures, screenshots and system images, each with where it is stored, its hash and who has access. This is what makes the record defensible.**

```
$ shasum -a 256 evidence/*.pcapng evidence/*.log > evidence/MANIFEST.sha256
```

9 **Confirm you understand what that hash proves. Put the same text into the Cryptography Toolkit twice and check the digest is identical, then change one character and check it is completely different. That property — and only that property — is what makes your evidence manifest tamper-evident.**

```
→ https://alfredang.github.io/cryptography-toolkit/
```


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Document a Security Incident to the Standard — Part 3/3

10 Record the notification decisions — who was informed, when, and which obligations were considered. In Singapore, consider whether the PDPA data-breach notification obligation to the PDPC and affected individuals is engaged, and record the reasoning either way.

11 **Ask your AI assistant to draft the incident narrative from your timeline.**

```
$ Prompt: 'Draft a factual incident narrative from this evidence timeline. Use only what the timeline supports. Mark every inference explicitly as an inference. Do not speculate about attribution.'
```

12 Check the draft line by line against the timeline. Delete every statement the evidence does not support and every attribution claim. Record what you removed — that edit is the professional skill this lab teaches.

13 Write the lessons-learned section: what detected it, what delayed the response, and the one control change that would have prevented or caught it earlier. Feed that change back into the Lab 6 plan.

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LAB 11 · VERIFY

Document a Security Incident to the Standard

✅ Expected result

Every timeline entry cites its evidence source, inferences are marked as inferences, the severity classification states the criteria used, and the evidence inventory carries a hash for every item. You can name at least one unsupported statement you removed from the AI draft.

IF IT DOESN'T WORK

The timeline has gaps

that is a finding about your logging, so record the gap rather than guessing what filled it.

Severity is contested

that means the criteria were not written down before the incident.

The AI names an attacker group

remove it; attribution needs intelligence you do not have.

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TOPIC 04 · HANDS-ON

Implement the Incident Response Configuration and Playbook

LAB 12

A9 · A7 · K4 — Implement configurations aligned to the incident response design: containment controls, an automated playbook and the human-approval boundary.

The final lab closes the loop. You take the incident from Lab 11 and implement the response as configuration: the containment controls that would stop it recurring, a documented SOAR-style playbook with explicit automation and approval steps, and detection rules derived from what you actually saw. You then test the playbook against a replayed scenario, measure the response time, and confirm that no automated action can cause more damage than the incident it responds to.

YOU'LL BUILD

Implemented containment configurations, a tested incident response playbook with a defined human-approval boundary, new detection rules, and a measured response time from detection to containment.

TOOLCHAIN

Firewall and host configuration, playbook template, Lab 8 detection rules, a generative AI assistant

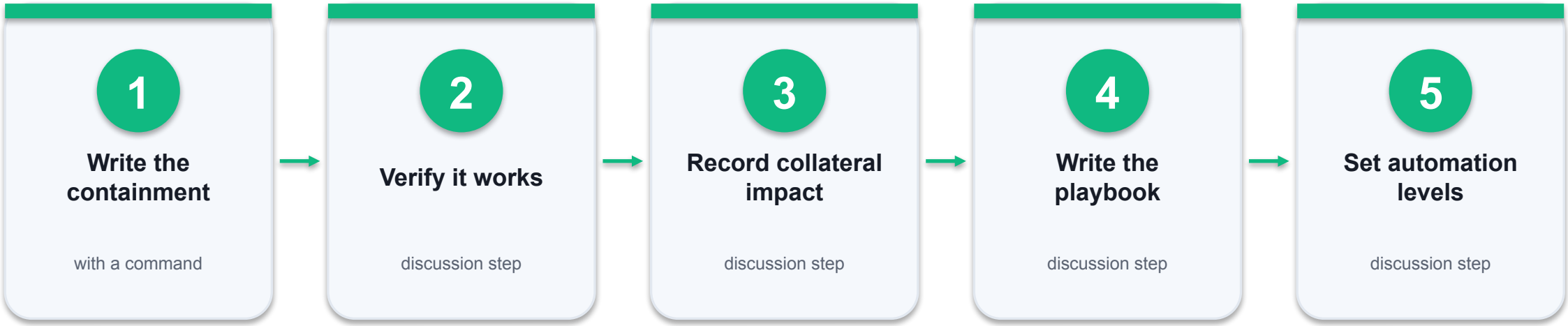
DONE WHEN

Every containment control has a verification command and a recorded output, every playbook step names an owner and an automation level, and you can state a measured detection-to-containment time against a target. No automated step can take a production service offline without a named approver.

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LAB 12 · PROCESS MAP

How Lab 12 Runs



THE POINT

A9 · A7 · K4 — Implement configurations aligned to the incident response design: containment controls, an automated playbook and the human-approval boundary.

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LAB 12 · PROCEDURE

Implement the Incident Response Configuration and Playbook — Part 1/3

1. **Write the containment configuration for the Lab 11 incident: the firewall rules, account actions and host isolation steps that stop this specific attack path, as commands rather than prose.**

```
$ iptables -I FORWARD 1 -s 192.168.56.101 -j DROP -m comment --comment 'INC-2026-001 containment'
```

2. Verify each containment control actually works, and record the verification command and its output. An untested containment control is an assumption.

3. Record the collateral impact of each containment action before it is ever needed. Isolating a host stops the attacker and also stops that host's business function — the responder must know which they are trading.

4. Write the playbook with the standard phases: Detect, Triage, Contain, Eradicate, Recover, Review. Each step names the trigger, the action, the owner and the expected duration.

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LAB 12 · PROCEDURE

Implement the Incident Response Configuration and Playbook — Part 2/3

5. Mark each step Automated, Automated-with-approval or Manual. Reversible, low-blast-radius actions can be automated; anything that takes production offline needs a named approver.

6. Define the escalation path with names and contact methods, and the time threshold that triggers each escalation. A playbook with no named person is a document, not a capability.

7. Add the detection rules from this incident to your Lab 8 ruleset so the same attack is caught earlier next time, and record the expected false-positive rate of each new rule.

8. Test the playbook against a replayed scenario. Walk every step, time it, and note where it stalls — the stall is where the playbook is wrong.

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LAB 12 · PROCEDURE

Implement the Incident Response Configuration and Playbook — Part 3/3

9 Measure detection-to-containment time and compare it against your target. If there is no target, set one now — a response time nobody is accountable for will drift.

10 **Ask your AI assistant to review the playbook for gaps and dangerous automation.**

```
$ Prompt: 'Review this incident response playbook. Identify missing steps, any automated action whose blast radius exceeds the incident it responds to, and any step with no named owner. For each finding, say what you would change.'
```

11 Fix every genuine gap. For any automated action flagged as too broad, either narrow its scope or move it behind human approval, and record the decision.

12 Feed the whole result back: update the Lab 6 risk management plan with the control changes and re-score the affected risks. The loop from incident back to plan is what makes this a management system rather than a set of documents.

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LAB 12 · VERIFY

Implement the Incident Response Configuration and Playbook

✅ Expected result

Every containment control has a verification command and a recorded output, every playbook step names an owner and an automation level, and you can state a measured detection-to-containment time against a target. No automated step can take a production service offline without a named approver.

IF IT DOESN'T WORK

The containment rule does not take effect

check the rule order; an earlier ACCEPT will win.

The playbook test stalls waiting for a decision

that is the finding, so add the decision authority.

The AI proposes automating everything

apply the blast-radius test yourself; that judgement is not delegable.

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TOPIC RECAP

Recap — Topic 04

1

Lab 10 — Centralise Logs and Triage Alerts with AI

Design the auditing arrangement that makes incident response possible, then use AI to reduce a flood of raw alerts into a ranked, workable queue.

2

Lab 11 — Document a Security Incident to the Standard

Produce an incident record that is complete enough to hand to a manager, an auditor or a regulator without a conversation attached.

3

Lab 12 — Implement the Incident Response Configuration and Playbook

Implement configurations aligned to the incident response design: containment controls, an automated playbook and the human-approval boundary.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1 · Threats & vulnerabilities

Mapped an ICT network and read its configuration, baselined normal traffic, trained an anomaly model, and produced a prioritised vulnerability worklist.

2

LO2 · Assets & risk plans

Built a classified asset register, scored a risk register with AI-challenged likelihoods, and produced a risk management plan mapped to the NIST CSF.

3

LO3 · Security measures

Designed a three-phase segmented architecture, implemented a default-deny ruleset with tuned detections, and hardened the AI itself against adversarial attack.

4

LO4 · Incidents & response

Specified logging, triaged alerts with AI, documented an incident to evidentiary standard, and implemented and tested a response playbook.

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NEXT STEPS

Continue Your Learning

1

NIST Cybersecurity Framework 2.0

https://www.nist.gov/cyberframework — the reference for Govern, Identify, Protect, Detect, Respond and Recover.

2

NIST AI 100-2 Adversarial ML

https://csrc.nist.gov/pubs/ai/100/2/e2025/final — the taxonomy of attacks against AI systems.

3

NSA/CISA Deploying AI Systems Securely

https://www.cisa.gov/ai — the joint guidance you applied in Lab 9.

4

MITRE ATLAS

https://atlas.mitre.org/ — adversary tactics and techniques targeting AI systems, with real case studies.

5

CISA Known Exploited Vulnerabilities

https://www.cisa.gov/known-exploited-vulnerabilities-catalog — the KEV list you used to prioritise.

6

Rebuild the labs

Redo each lab on your own network with written authorisation, until the workflow is automatic.

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Practice Exam — Test Yourself Before the Assessment

🎯 https://exams.tertiaryinfotech.com

Open the practice exam portal

Go to exams.tertiaryinfotech.com and find the Network Security practice exam.

Attempt it before the assessment

Sit it under exam conditions — closed book, timed — so the result means something.

Review every wrong answer

Trace each one back to the slide and Learner Guide section that covers it.

Re-attempt until consistent

Repeat until you are scoring consistently, not just once.

It mirrors the knowledge questions

The practice questions rehearse the same K1–K4 knowledge the WA-SAQ assesses.

It does not replace the labs

The Case Study assesses ability — for that, rebuild the labs, don't re-read them.

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CONTINUE WITH TERTIARY INFOTECH

Recommended Courses

1

WSQ - Network Securities for Beginners

Build the underlying networking and security foundations if you want more depth on the fundamentals.

2

WSQ - Ethical Hacking and Penetration Testing

Learn to test your own defences the way an attacker would.

3

WSQ - Machine Learning with Python

Go deeper on the models behind the anomaly detection you built in Labs 2 and 9.

4

WSQ - Cybersecurity Incident Response

Extend Topic 4 into a full incident-response and digital-forensics capability.

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WE'RE HERE TO HELP · DURING AND AFTER THE CLASS

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Practice exam
exams.tertiaryinfotech.com

6

Course labs
github.com/tertiarycourses/TGS-2024051414-AI-for-Network-Security

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Assessment

1

Written Assessment (WA-SAQ)

1 hour · open book · short-answer knowledge questions covering K1–K4.

2

Case Study (CS)

1 hour · open book · a realistic organisational scenario covering A1–A9.

3

Digital attendance

Remember to take the Assessment digital attendance (TRAQOM) before you start.

4

Submit on the LMS

Upload your completed answers at https://lms-tms.tertiaryinfotech.com/

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ON ASSESSMENT DAY

Assessment Flow



REMEMBER

All five steps are mandatory for WSQ funding — missing the digital attendance or the TRAQOM survey can invalidate your claim.

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Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for every WSQ-funded course.

2

Trainer shows the QR

The trainer or administrator displays the digital attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum of 75% attendance is required to be eligible for assessment and funding.

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STAY SECURE

Thank You!

You can now apply AI to network threat analysis, risk management, security design and incident response — and defend the AI itself.

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